



An Introduction to AI & where Computer Vision Meets Creation

13 October
2025

v. 2.1



Seminar mixed with theory & practical sessions



Theory Journey – How the course builds conceptually

From Vision to Intelligence:

How machines learn to see

Recognition & Tracking:

Teaching machines to follow what they see

Deep Learning for Vision:

Neural networks that understand images

Generative & Responsible Vision:

When machines start to create — and deceive

Applied & Creative Vision: **Designing with AI & looking ahead**

The Practical Journey – Creating the Pixel Machine Lab

Image Manipulation with OpenCV · Mini Image Gallery

Face & Object Tracking · Dataset Annotation

Training Custom Models · Gesture Recognition · YOLOv9

Micro Deep Dives:
Generative Vision · Diffusion · Ethics · Bias · AI Tools

Spatial AI and 3D Point Cloud Exploration with Open3D

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How Computer Vision works

Face Detection & AI Bias

Practical Tasks

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Definition of Computer Vision

- Importance and Use Cases of Computer Vision
- Computer Vision vs. Image Processing,

What is it, Why is it Important, Where is it used, Basic Techniques & Key Concepts

Image Library, Face Detection



Machine Perception (Physical AI) technologies are built to mimic human senses

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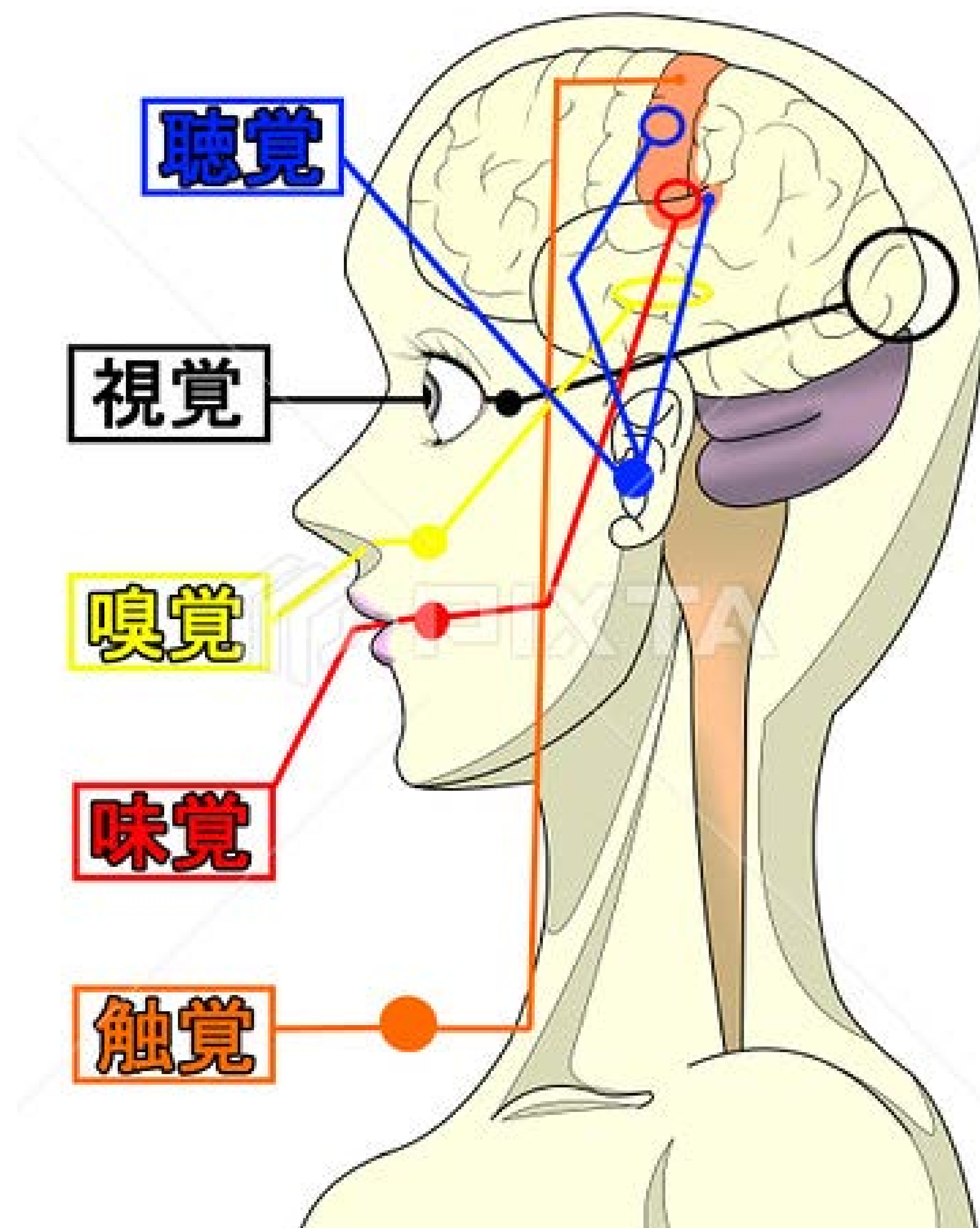




Machines that See, Hear, Feel, Smell, and Taste

- | | |
|--------------|-------------------|
| 1. Visual | 5. Tactile |
| 2. Auditory | 6. Vestibular |
| 3. Gustatory | 7. Proprioception |
| 4. Olfactory | 8. Interoception |

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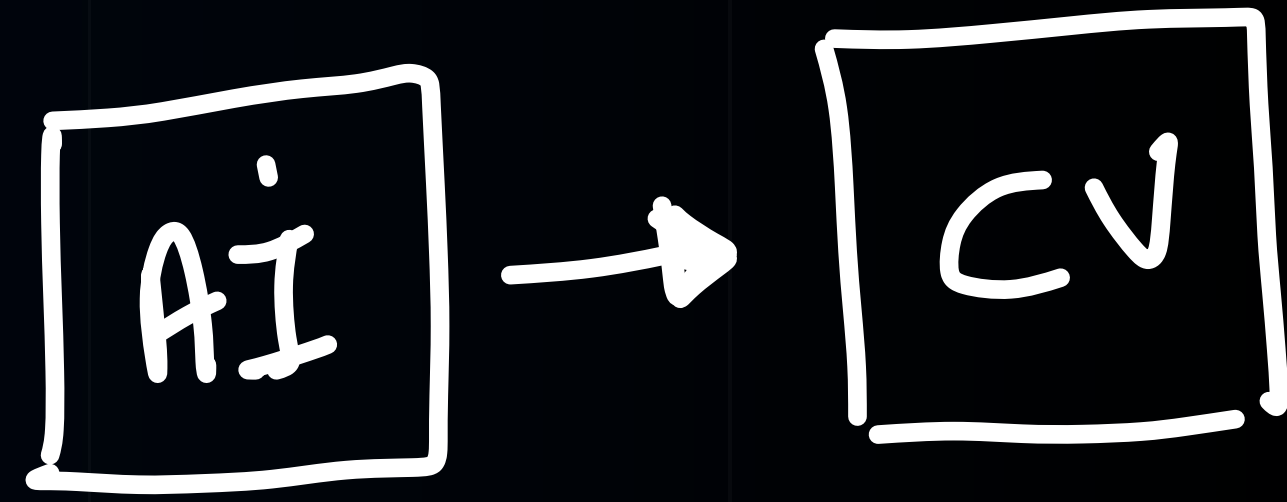




Introduction to Computer Vision

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Computer Vision (CV) **Trains Computers to Interpret &** **understand The Visual World.**

Computer Vision (CV) can be your way
in to AI as Designers & Front-End
Developers – Just try it out!

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The Barcode Scanner as an example for CV





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90%

of data is
processed
by human in
visual form



Computer Vision has a huge impact on business - increase efficiency and cut costs, freeing up teams to carry out more creative tasks.

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Question

Which tasks and real-world scenarios of CV have you experienced?

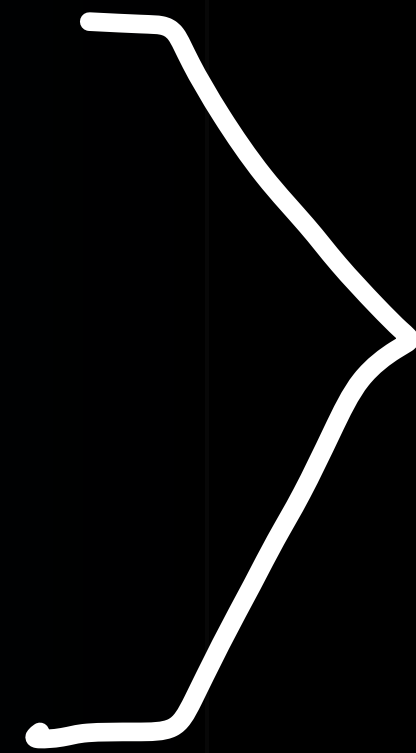
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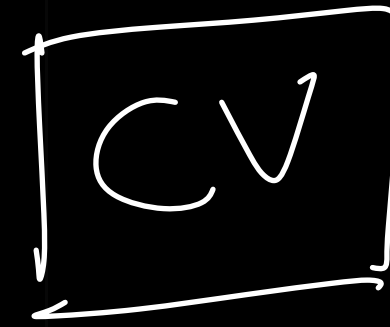


Tasks based on computer vision

- Traffic inspectors
- MRI scan-analysis experts
- sports umpires
- checkout assistants
- truck drivers
- equipment inspectors

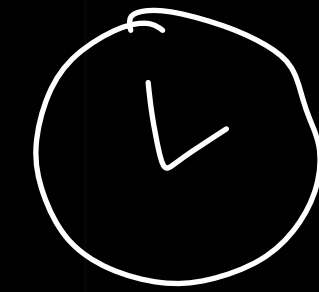


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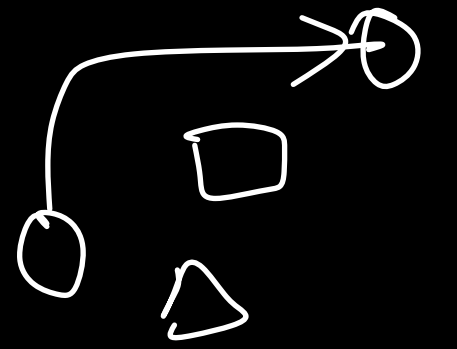
AUTOMATED

=



FASTER

+



EFFICIENT





Benefits of automating tasks with CV

1. Minimizes costs
2. Reduces errors
3. Increases processing speeds
4. Obliterates days off sick days
5. Frees human employees to carry out more meaningful tasks

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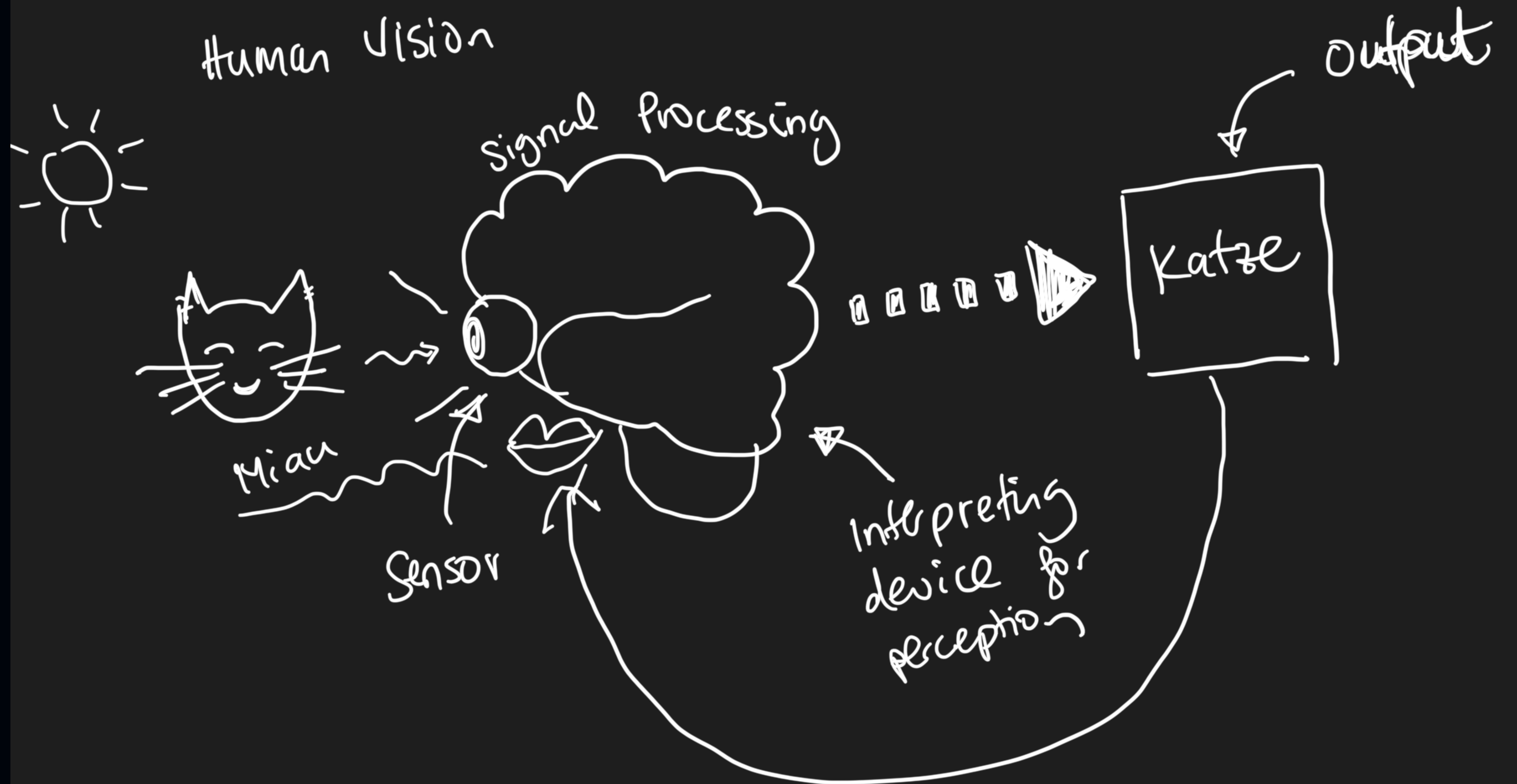




How does Computer **Vision work?**

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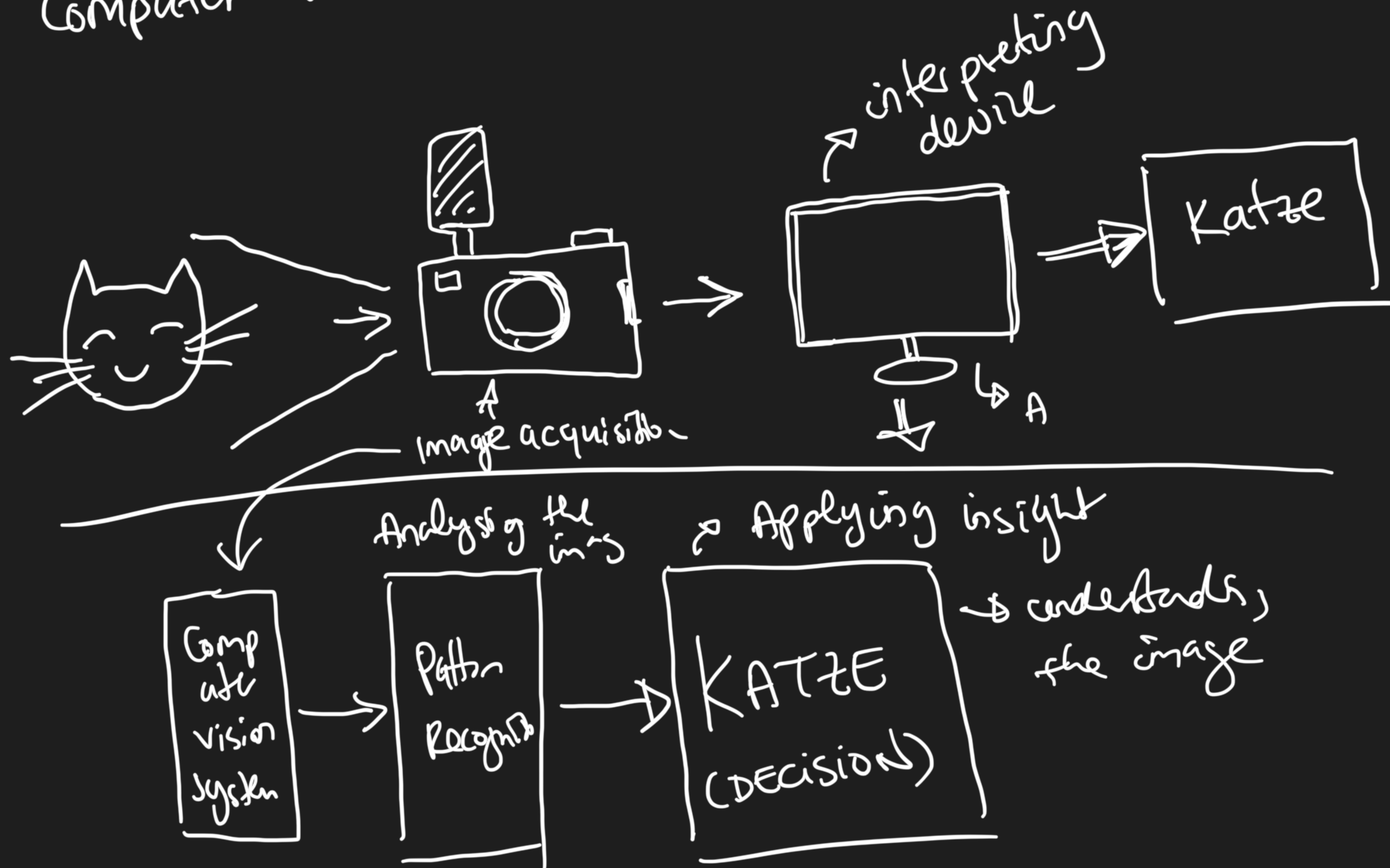


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Computer Vision



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There are 2 types of CV algorithms:

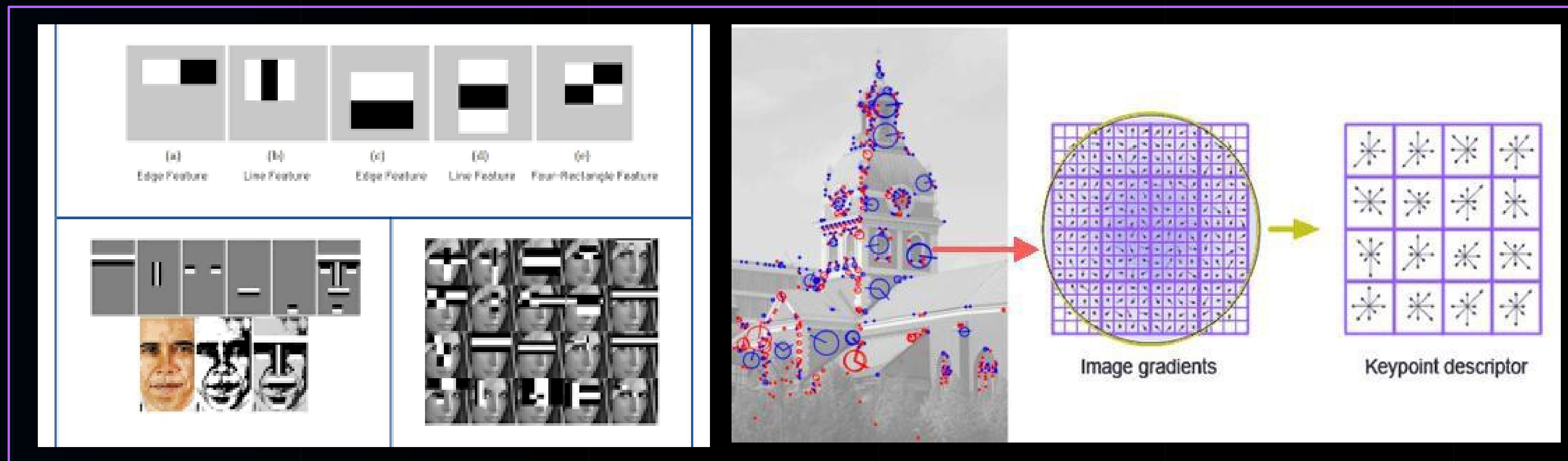
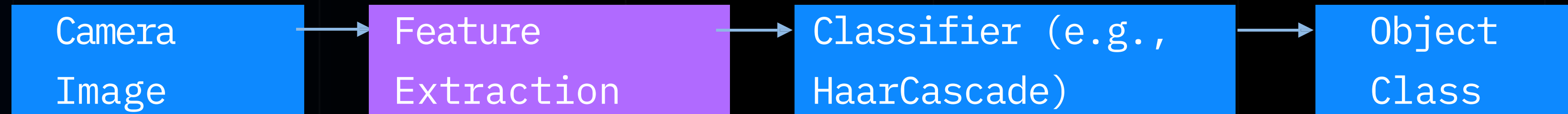
Data-Driven Approach
(Classical) vs. Deep Learning
Approach in Training
Computer Vision Models

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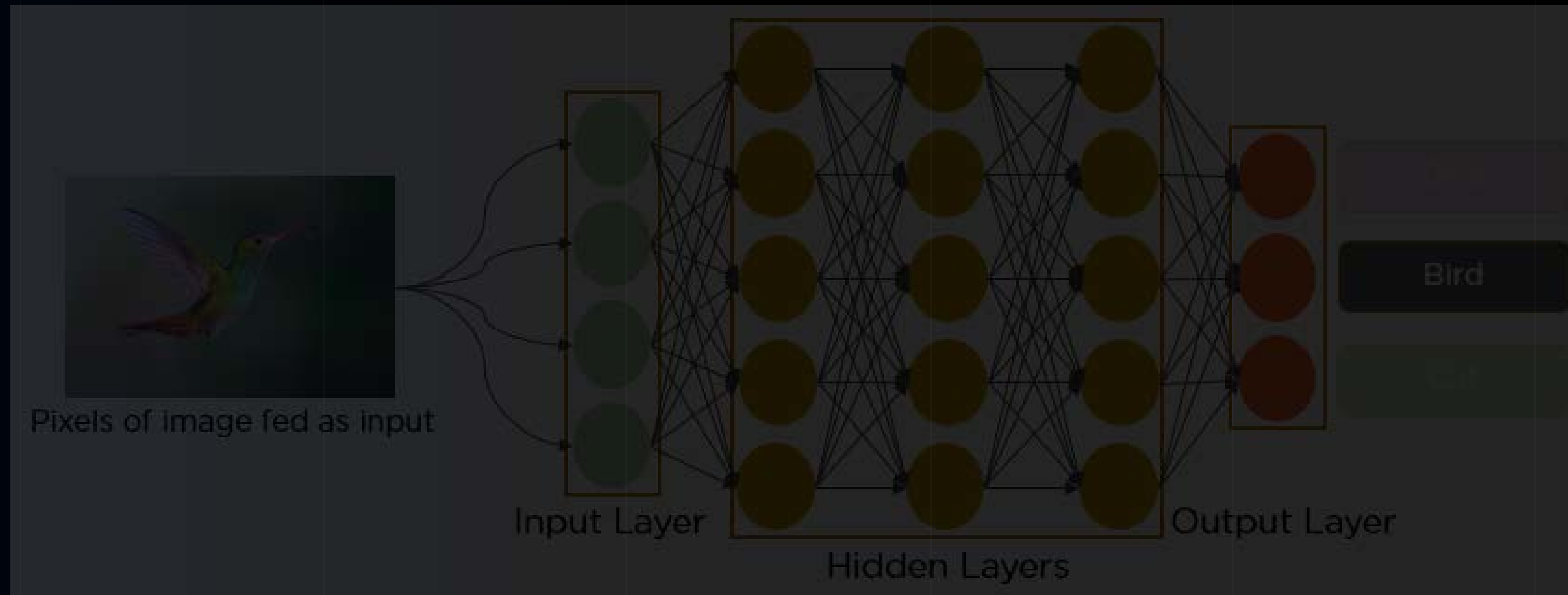
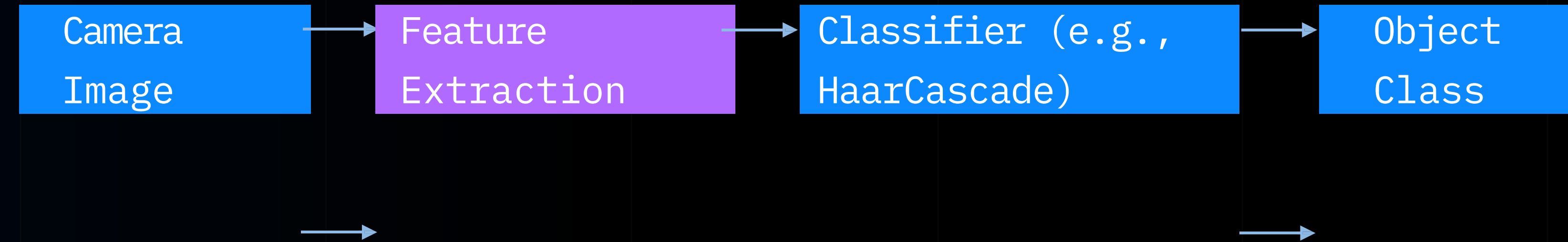


Computer Vision Pipeline – Classical CV / Data-Driven Approach





Deep Learning Computer Vision Pipeline



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Computer Vision Tasks

Classification



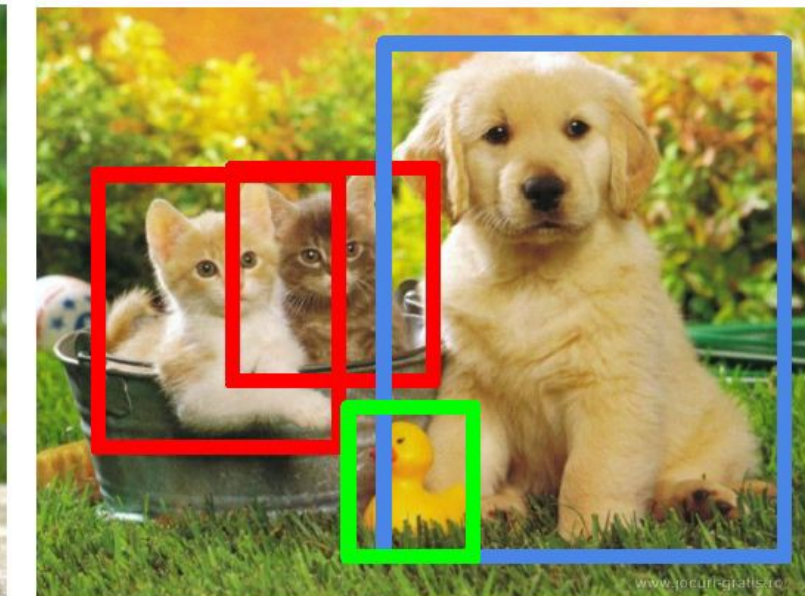
CAT

**Classification
+ Localization**



CAT

Object Detection



CAT, DOG, DUCK

Instance Segmentation



CAT, DOG, DUCK

Single object

Multiple objects





Computer Vision Tasks

Image Classification assigns a class label to an entire image or photograph. It's the task that the computer classifies the entire input image into one of the predefined labels or classes

Classification + Localisation extends image classification by identifying *where* in the image the classified object is located. The algorithm predicts both the class label and the coordinates of a bounding box around the object.

Object Detection involves identifying what is present in an image and determining where in the image the objects are. It is typically achieved by using bounding boxes

Instance Segmentation partitions an image into different regions, each corresponding to different object classes. There are two types: Semantic Segmentation (all pixels of the same object have the same label) and Instance Segmentation (different objects of the same class have different labels)."





Feature generation for Computer Vision Systems

Image classification
Object detection
Image Segmentation
Object Tracking
Image generation
Edge detection
Face Recognition
OCR (Optical Character
recognition)
Pattern detection
Feature matching

...

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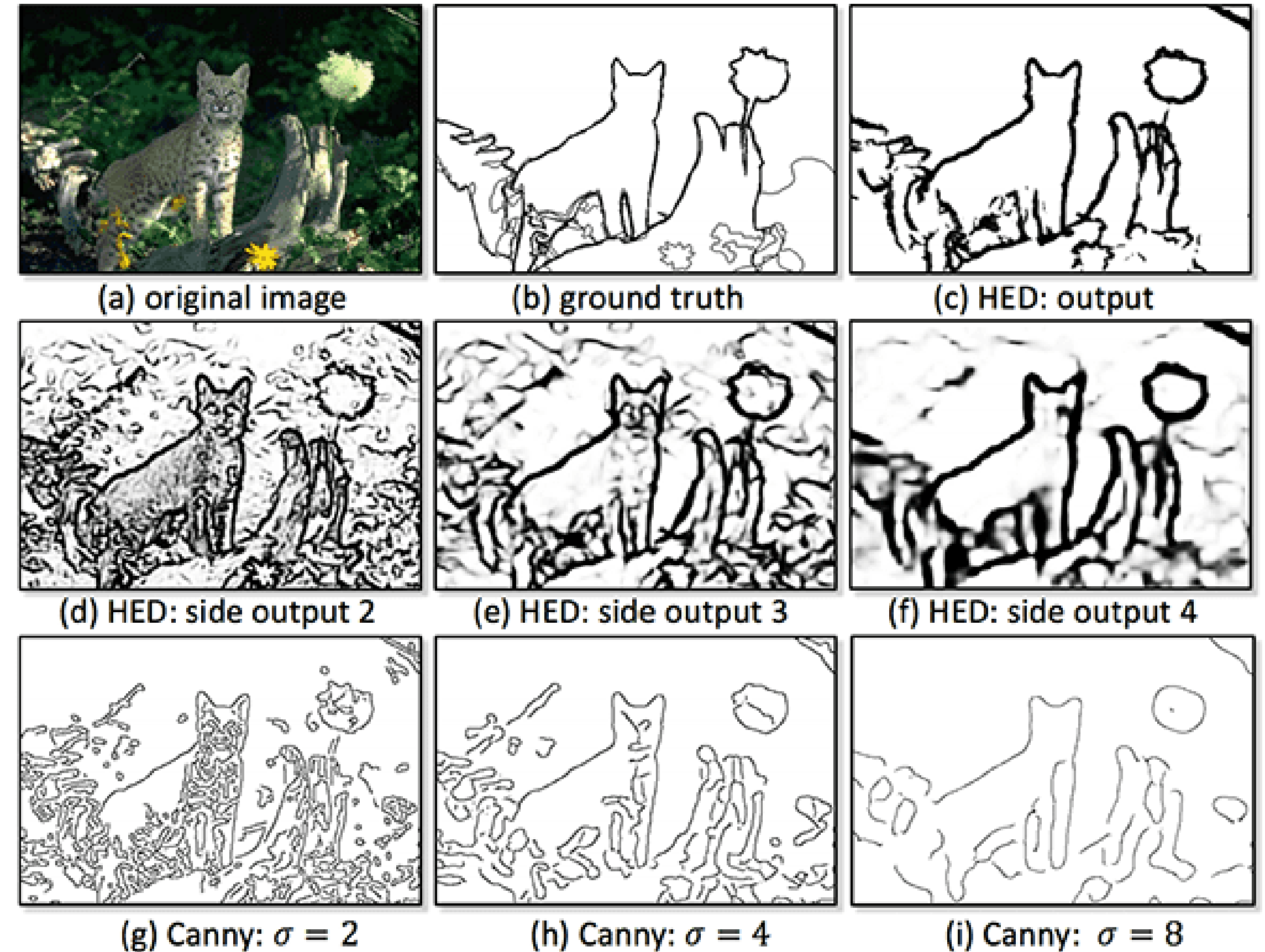




Edge Detection

Edge Detection is a fundamental tool in image processing, machine vision and computer vision, particularly in the areas of feature detection and feature extraction, used to identify points in a digital image where the image brightness changes sharply.

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Optical Character Recognition

OCR technology can recognize and extract text from images, videos, or even a live camera stream.

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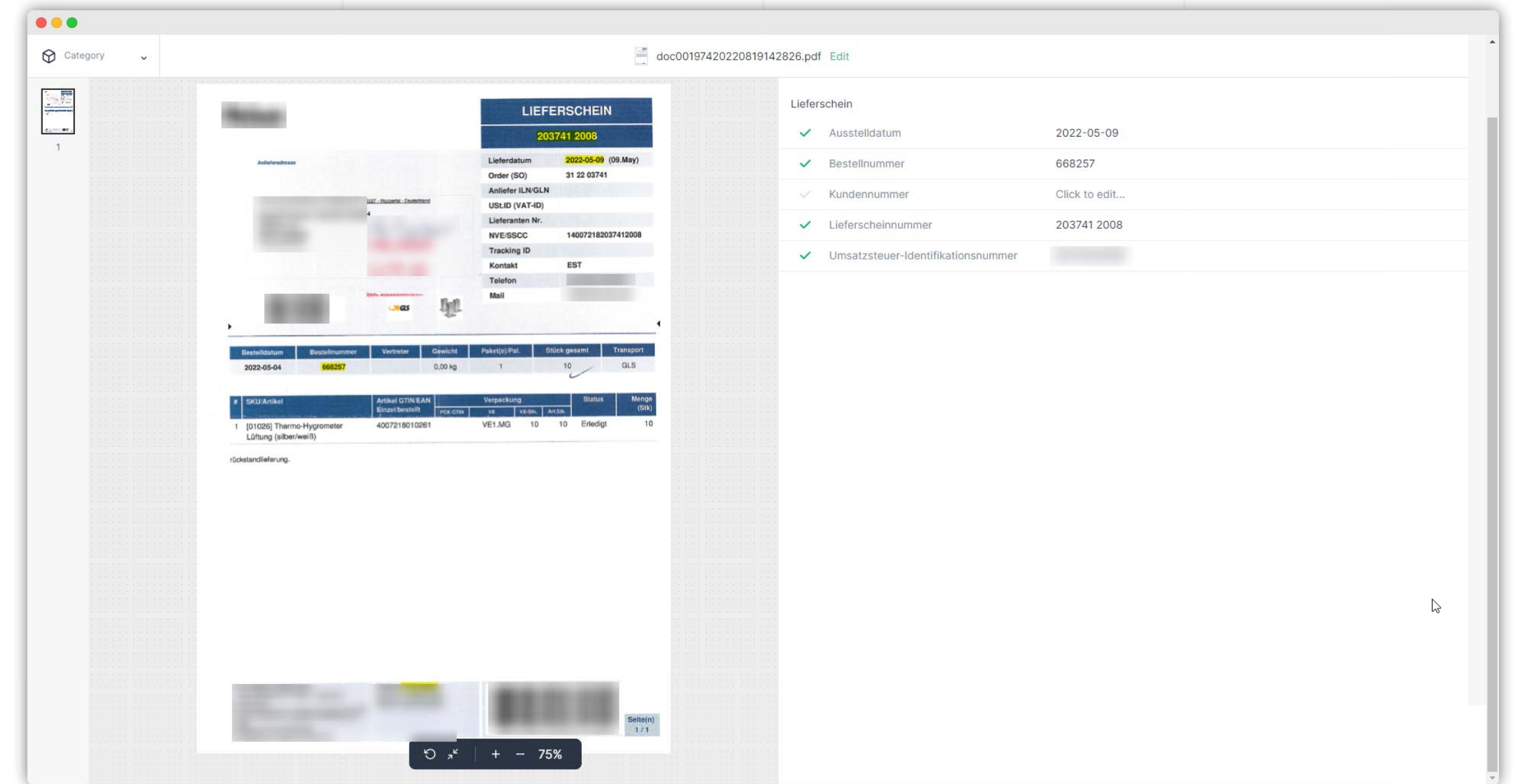
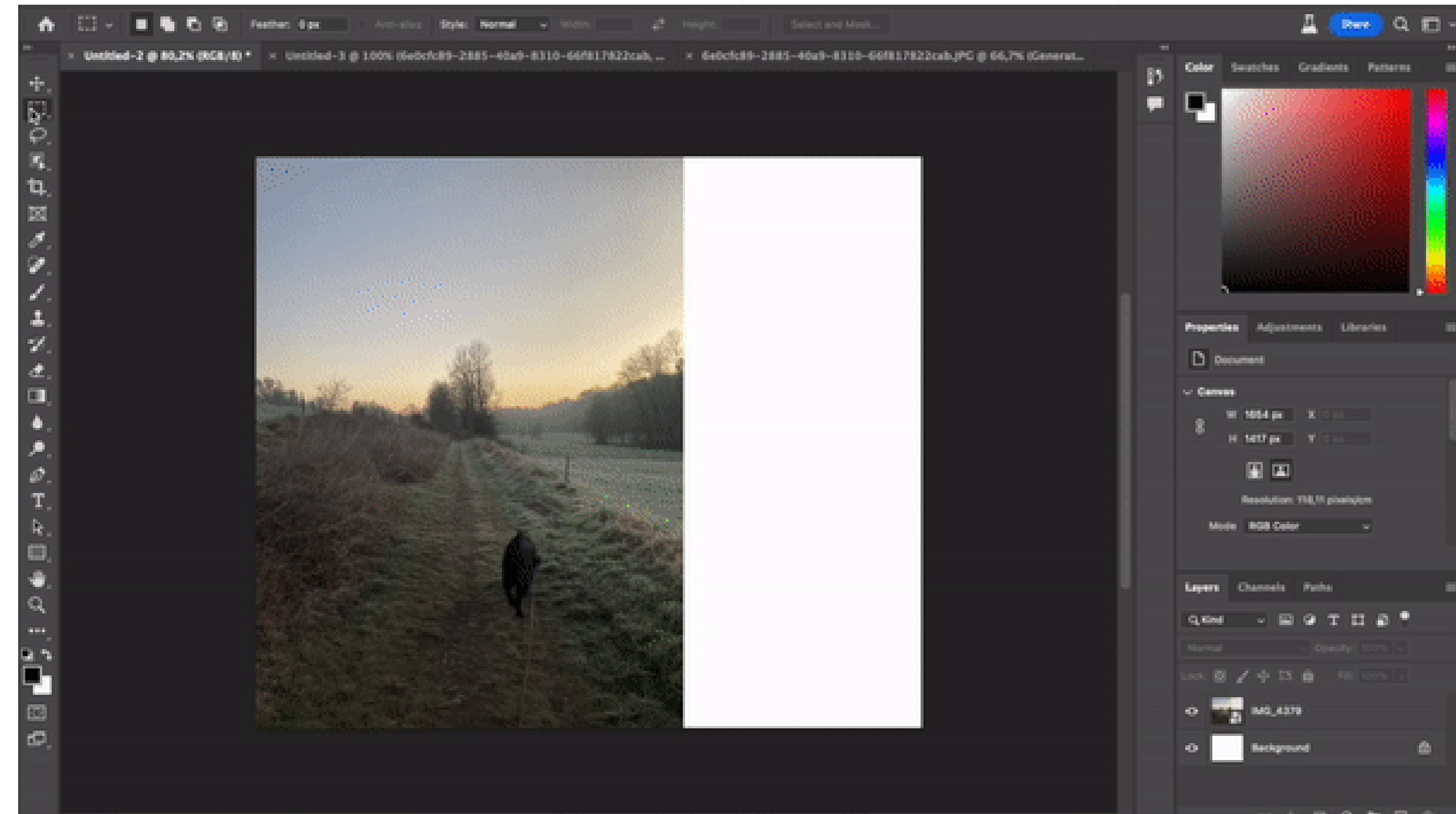




Image Generation

Image Generation techniques like GANs (Generative Adversarial Networks) can create new images that resemble a given dataset. They have been used to generate realistic human faces, change image style, and even create art.

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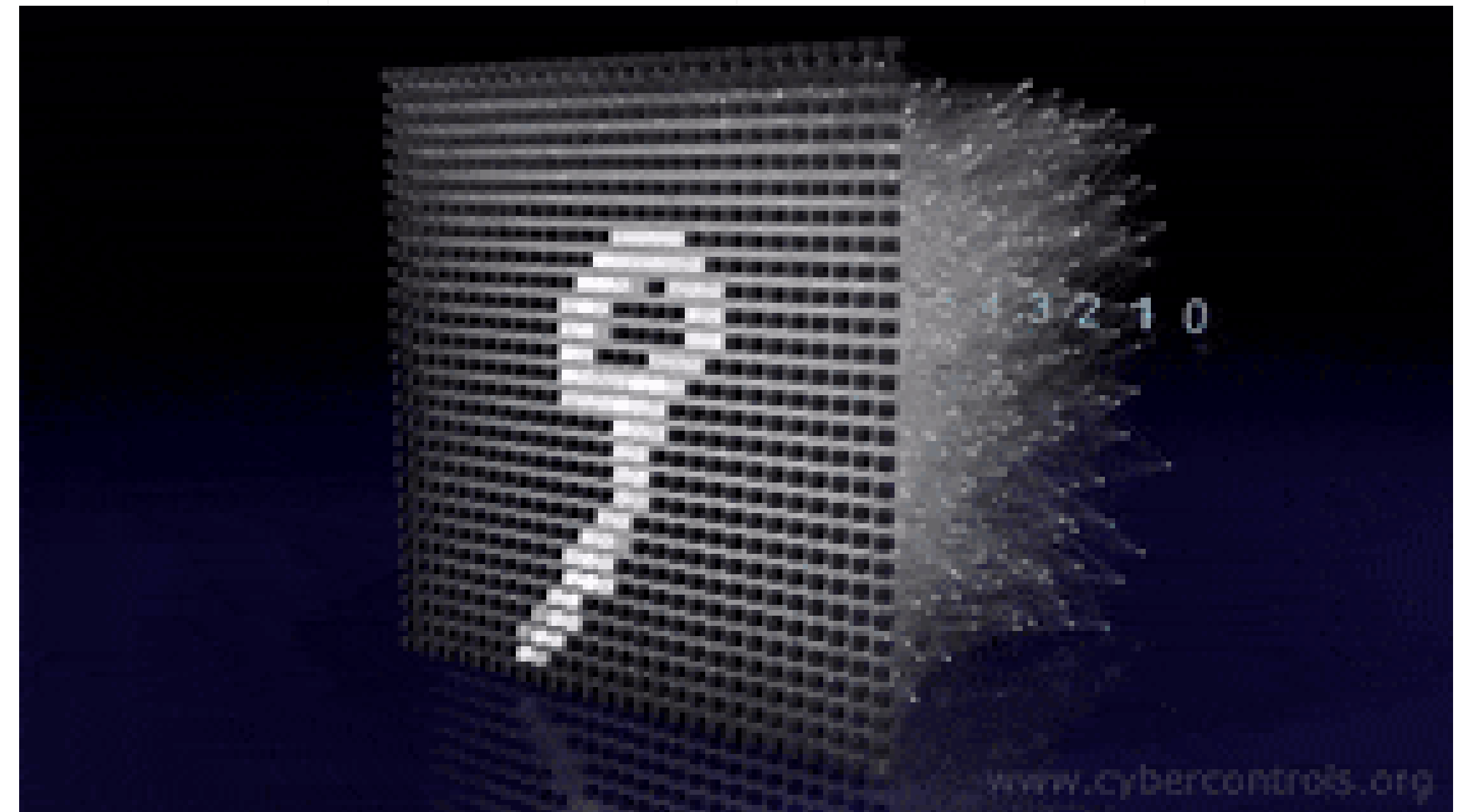




Pattern Recognition

Pattern Detection involves identifying and locating regularities in data. This includes simple methods such as template matching, more complex methods such as Hough Transform, or even machine learning based methods.

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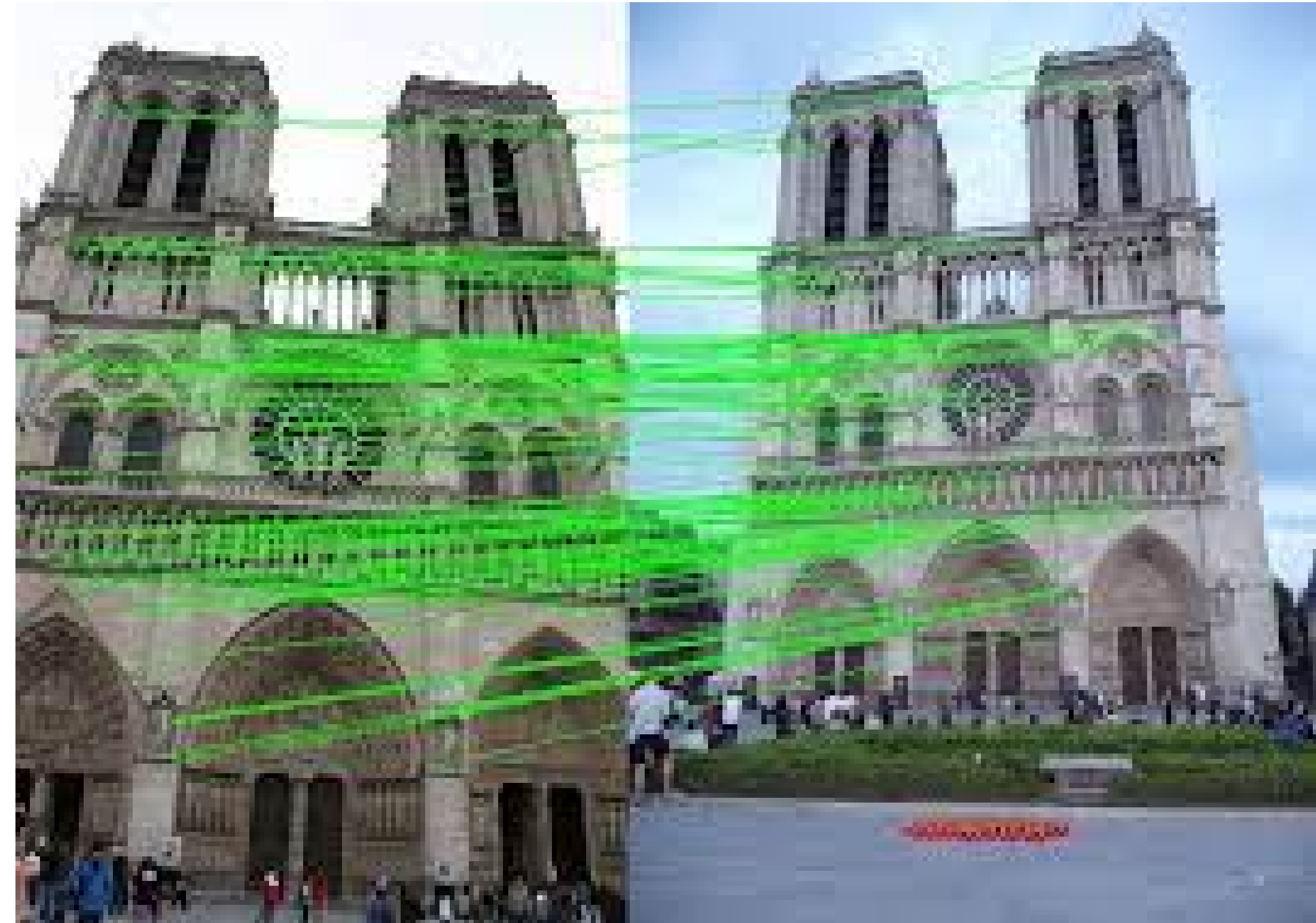


Feature Matching

Find similar characteristics and groups them together, e.g similar objects in photographs

This is also a basic localization technique

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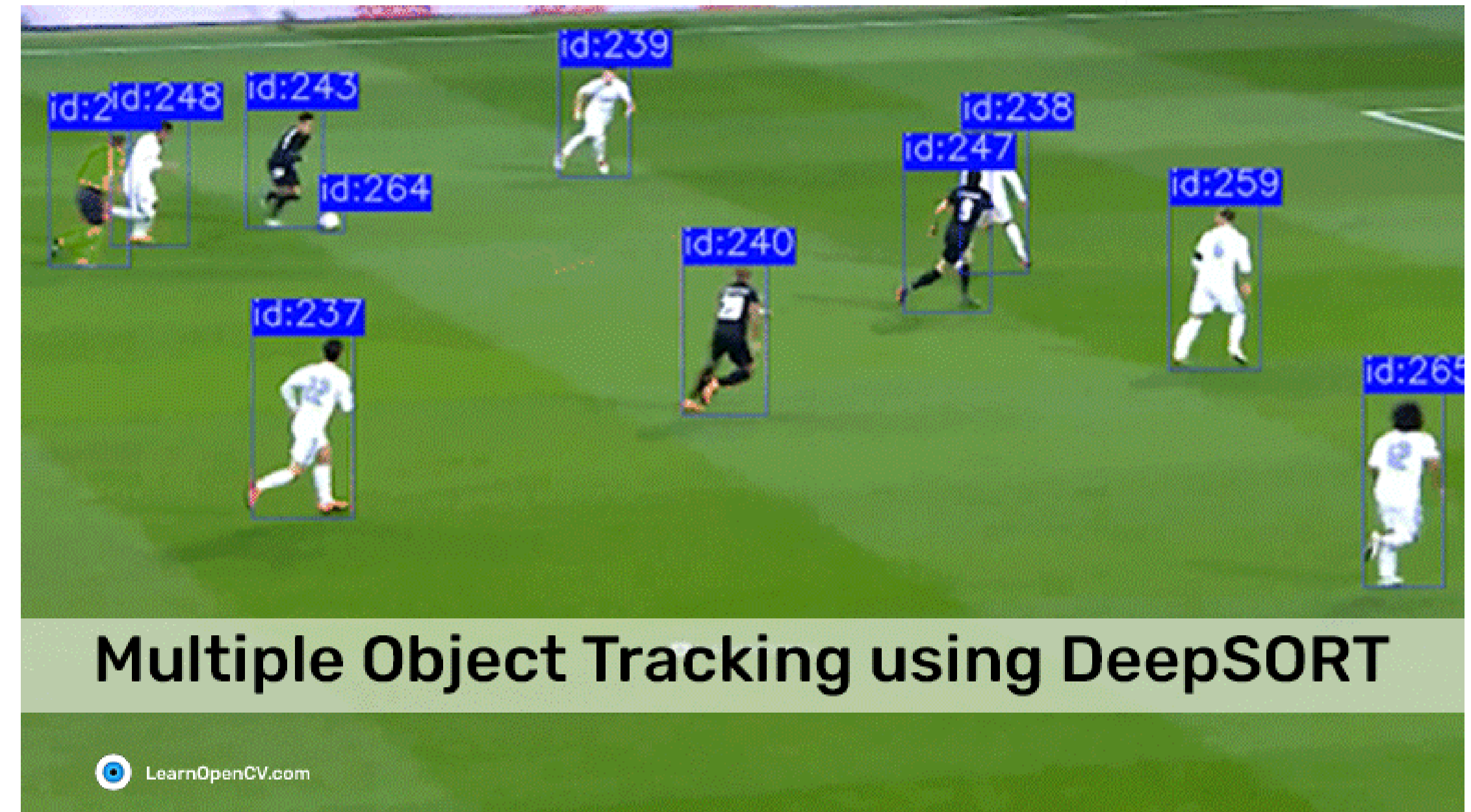




Object Tracking

Object Tracking is the process of locating a moving object over time in a video stream. This is typically achieved by using algorithms that take into consideration both the previous and current frames

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Levels of Vision

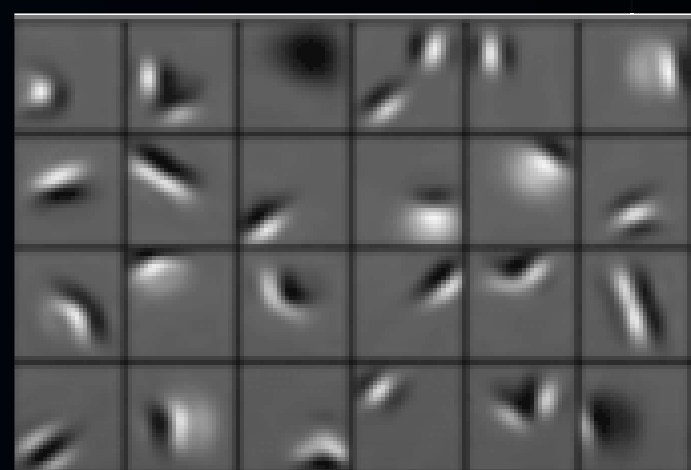
From Pixels to Meaning

Low-Level Vision

Focus: Raw pixel patterns
Finds edges, corners, textures
Techniques: Edge Detection, Pattern Detection, Feature Matching

Feature Type: simple, geometric

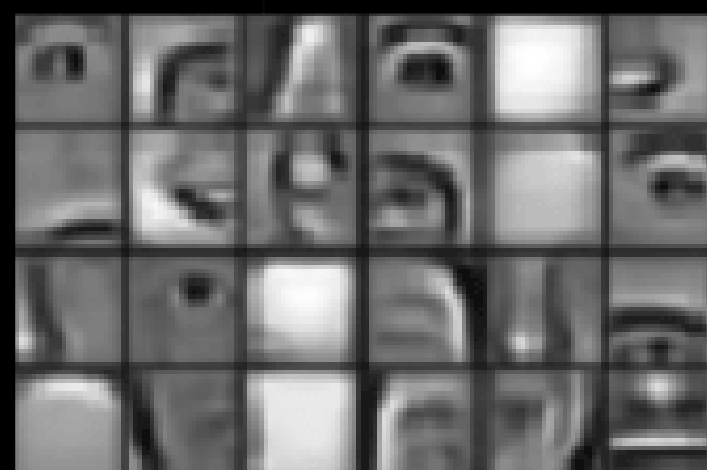
e.g. Edges, dark spots



Mid-Level Vision

Focus: Objects and motion
Groups low-level features into shapes or regions
Techniques: Object Detection, Segmentation, Tracking
Feature Type: shapes, boundaries, position

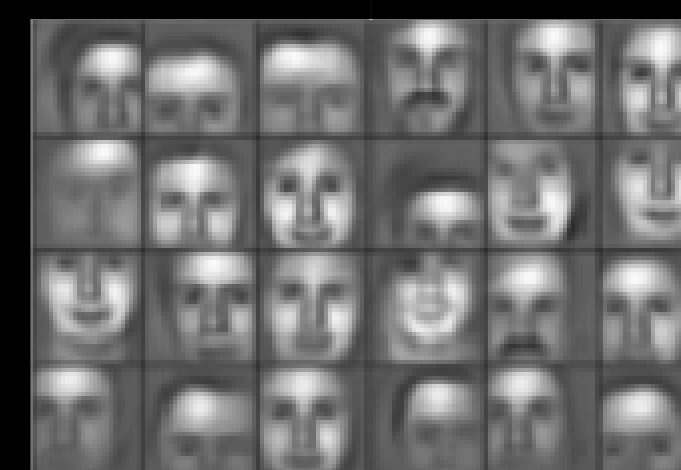
e.g. Eyes, ears, nose



High-Level Vision

Focus: Meaning and identity
Interprets objects in context
Techniques: Face Recognition, OCR, Image Classification, Image Generation
Feature Type: semantic, abstract

e.G. Facial structure



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From pixels → edges → objects → meaning



Practical Task

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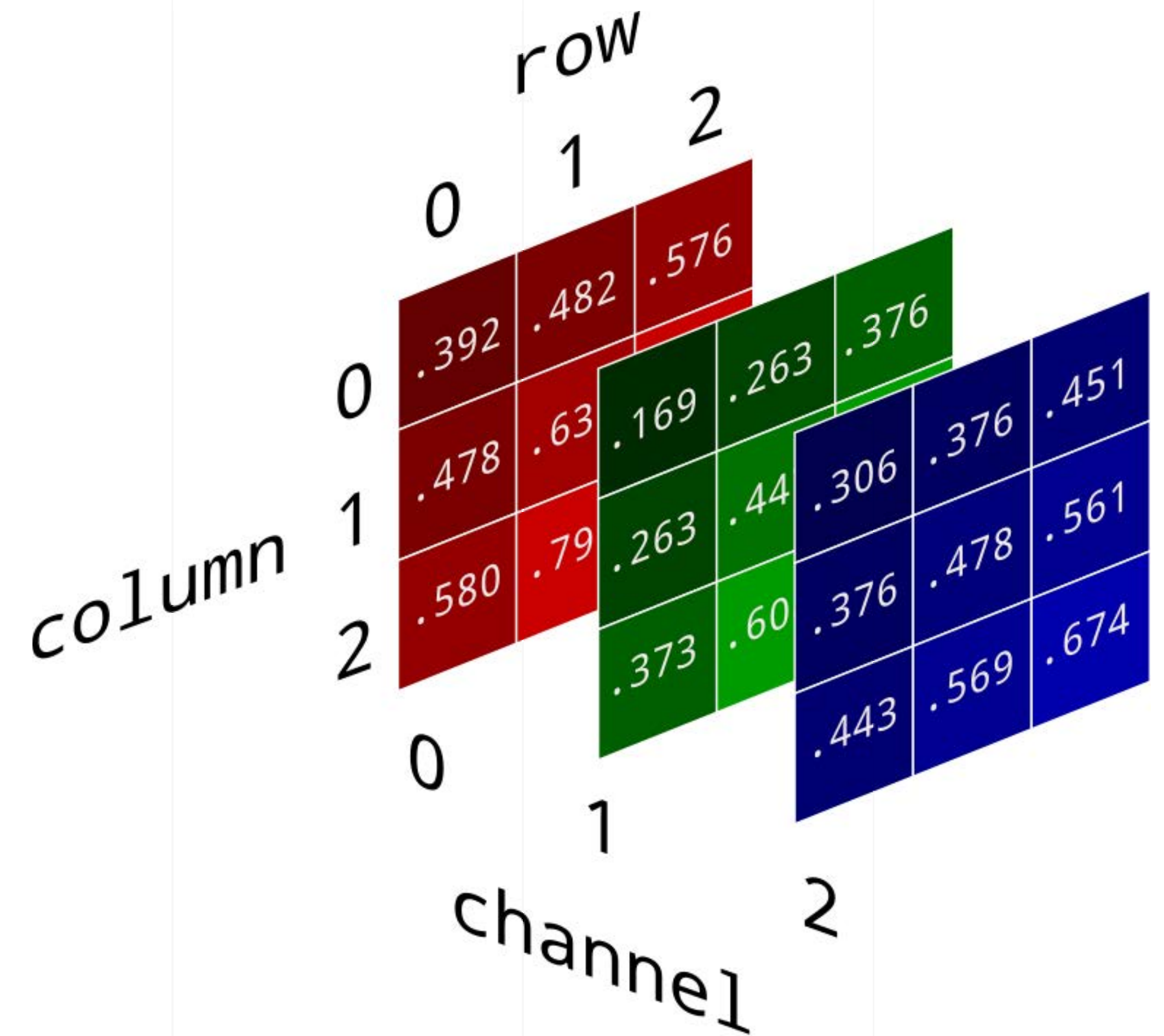




An image is an array or matrix, consisting of 3 channels and rows + columns

For greatest flexibility we can pass Images to numpy arrays to do scientific operations on it.

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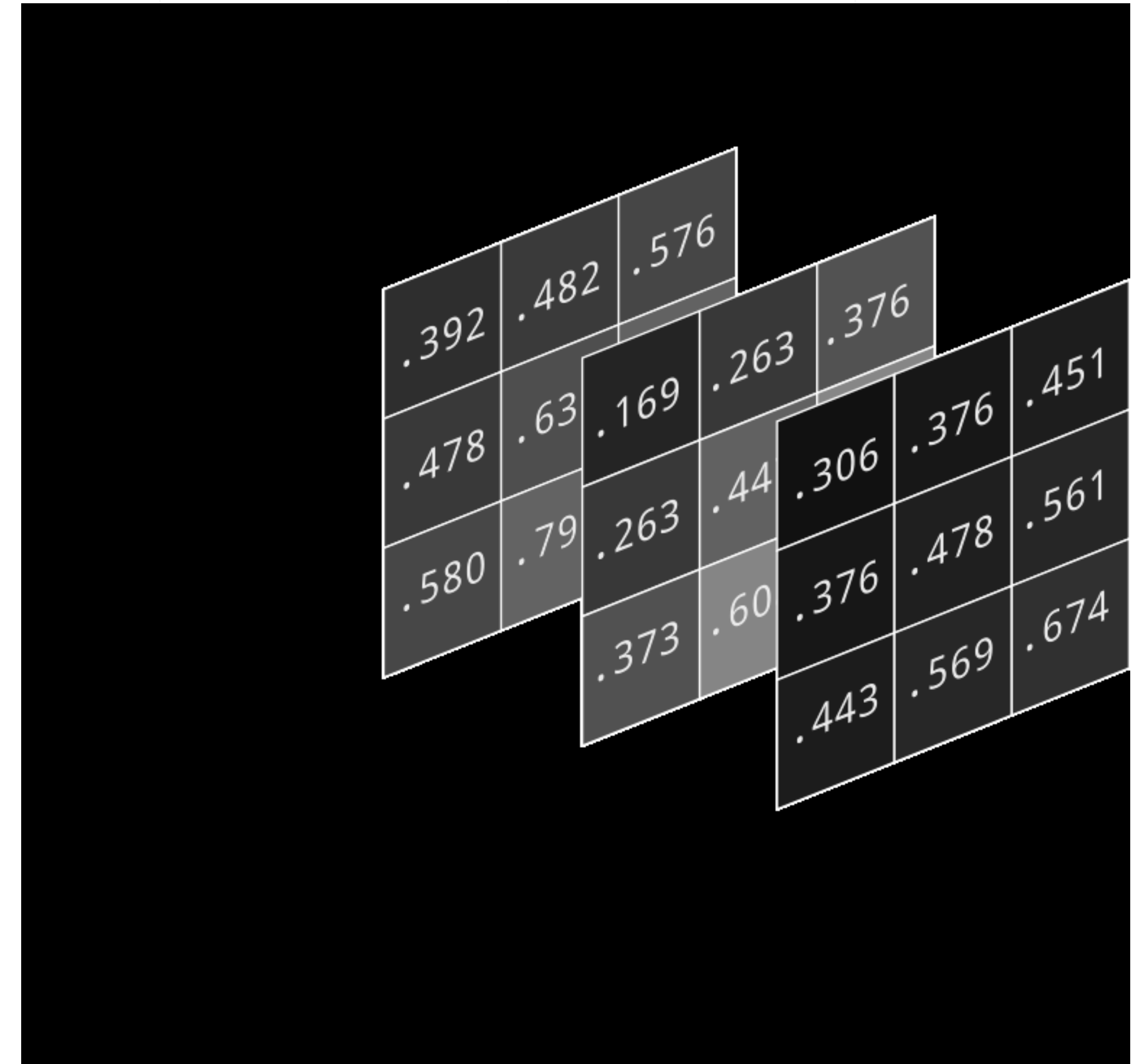




Machines are Color-Blind to process images faster.

First step in all algorithms is to make an image
greyscale so that the algorithm only deals
with 1 channel instead of 3

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Numpy_Image Basics

```
import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
import cv2

myList = [1,2,4,5,10]
arr = np.array(myList)

arr = np.arange(0,10)

np.arange(0,10,2)

np.zeros((5,5))

arr = np.ones((2,4))

np.random.seed(101) # watch video for details
arr = np.random.randint(0,100,10)

arr2 = np.random.randint(0,100,10)

arr.max()

arr.min()

arr.mean()

arr.argmin()

arr.argmax()

arr.reshape(2,5)
```

```
# Indexing
mat = np.arange(0,100).reshape(10,10)

print(mat)

row = 0
col = 1

mat[row,col]

# With Slices
mat[:,col]

mat[row,:]

mat[0:3,0:3]
mat.max()

print(mat)

#matplotlib inline
pic = Image.open('img/red_panda.jpeg')
pic

type(pic) #Check the type of the image object
pic_arr = np.asarray(pic)
pic_arr.shape
pic_red = pic_arr.copy() # create copy to not change the original
pic_red[:, :, 1] = 0 # Zero out contribution from green
pic_red[:, :, 0] = 0 # Zero out contribution from blue
#plt.imshow(pic_red)
#plt.show()
```

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Image processing tasks

```
import cv2
#main
from image_processing import load_image, display_image,
save_image, translate_image, rotate_image,
convert_to_gray, gaussian_blur, draw_line
# from sound_processing import load_sound, play_sound

imagePath = 'img/color_wheel.png'
# Use the image processing functions
img = load_image(imagePath)
img = cv2.imread(imagePath)

# Translate, rotate and resize image
translated_img = translate_image(img, 200, 50)
translated_img = rotate_image(translated_img, 90)
translated_img = convert_to_gray(translated_img)
translated_img = gaussian_blur(translated_img,
kernel_size=(25,25))

translated_img = draw_line(translated_img,
color=(255,255,0))

display_image(translated_img)

# Copy image into a new file
imgCopy = save_image(img, imagePath)
display_image(imgCopy)

# create a new image
```

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Basic Image Processing: Channel Manipulation & Image Gallery

Computer Vision Basics Aufgabe

🎯 Ziel:

Verstehen, wie digitale Bilder aufgebaut sind – durch das Analysieren von Farbkanälen, Graustufen und Filtern in OpenCV.

🧩 Aufgabe:

1. Wählt ein Farbbild aus eurem Datensatz.
2. Zeigt das **Originalbild** an.
3. Trennt die drei Farbkanäle (Rot, Grün, Blau) und zeigt sie einzeln an.
4. Wandelt das Bild in **Graustufen** um.
5. Erstellt eine **geblurte Version** (z. B. Gaussian Blur).
6. Erstellt eine **gefilterte Version** (z. B. Canny Edge Detection).
7. Zeigt alle Versionen nebeneinander in eurer eigenen **Image Gallery**.
8. Vergleicht die **Pixelwerte** – wie verändern sich sie zwischen Original, Gray und Blur?

💬 Bitte postet als Antwort auf diesen Thread:

- 🖼️ Einen Screenshot eurer „Image Gallery“
- 💡 Eine kurze Beobachtung: Wie verändern sich Schärfe und Pixelwerte zwischen den Varianten?
- ⚙️ Optional: Testet zusätzlich einen anderen Farbraum (z. B. HSV oder LAB)

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Face Analysis & Recognition

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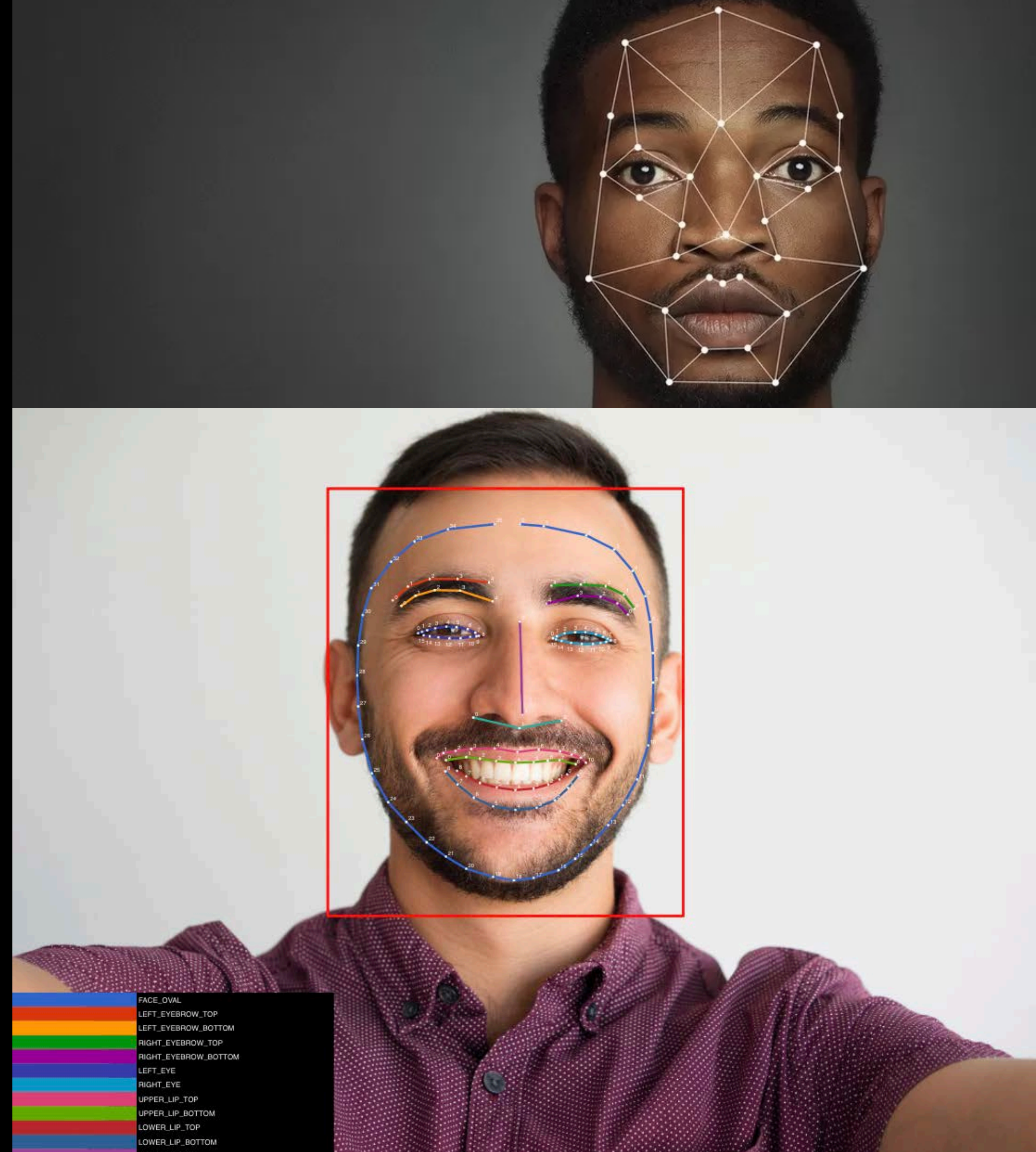




Face Recognition

Face Recognition technology can identify a person from a digital image or a video frame.

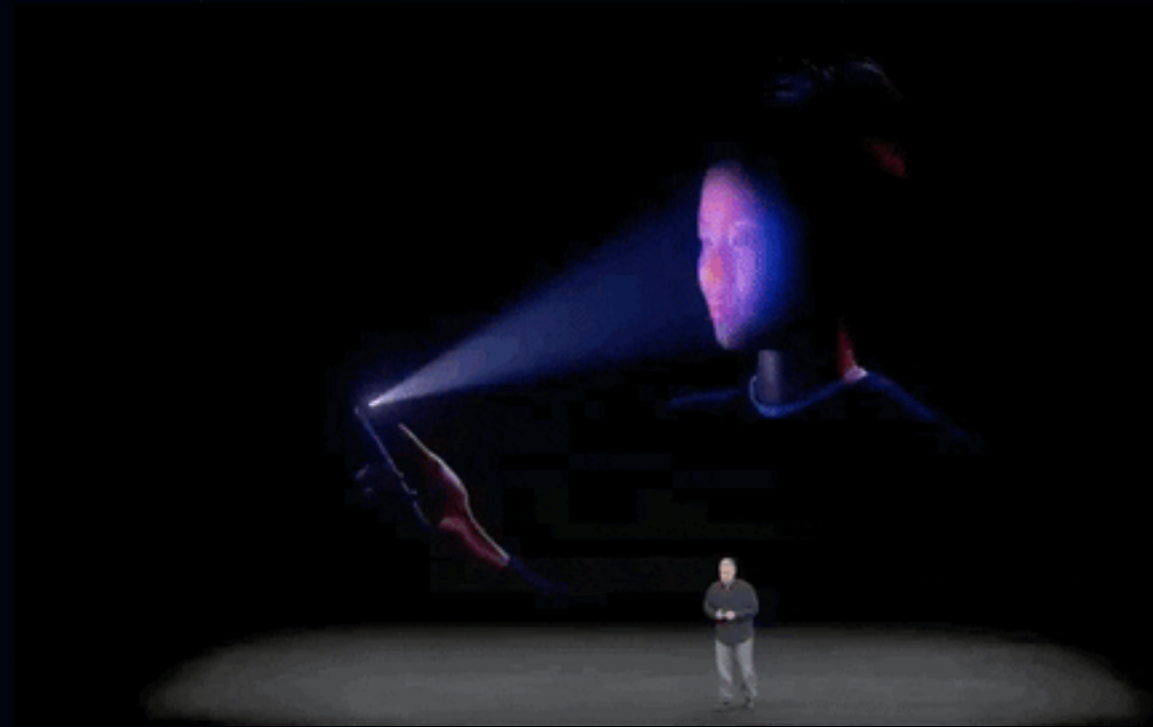
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Wide application of CV for human faces

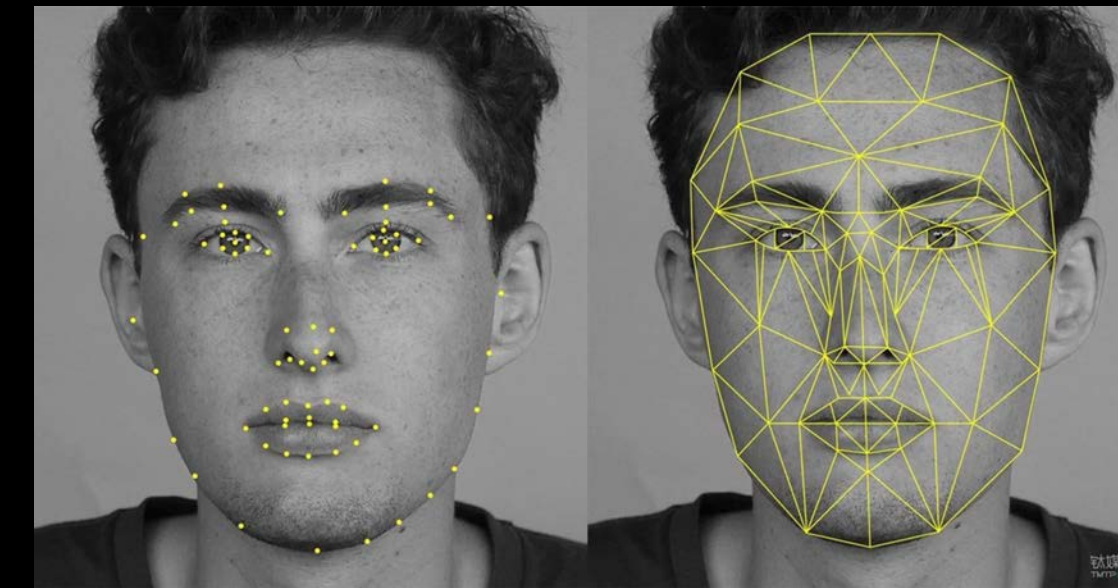
Understanding human faces is fundamental for a variety of applications. It's not just about recognizing who the person is, but also about understanding their expressions, estimating their age, and even tracking their movements.



Visual Authorization & Identification



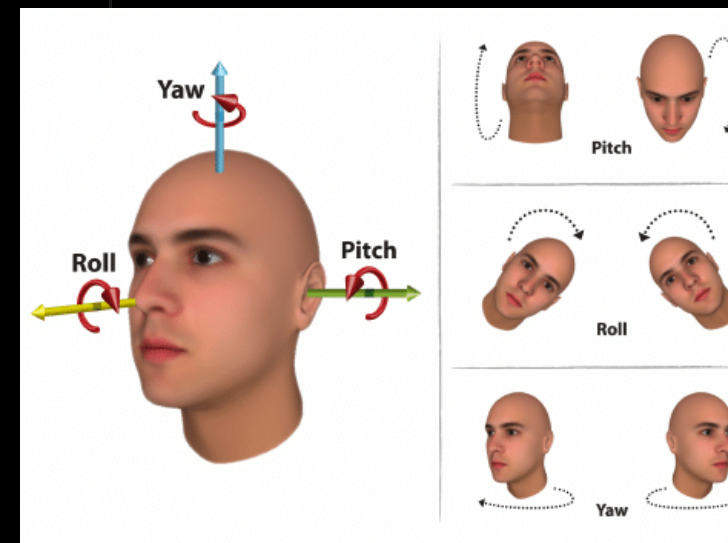
Visual surveillance & multimedia



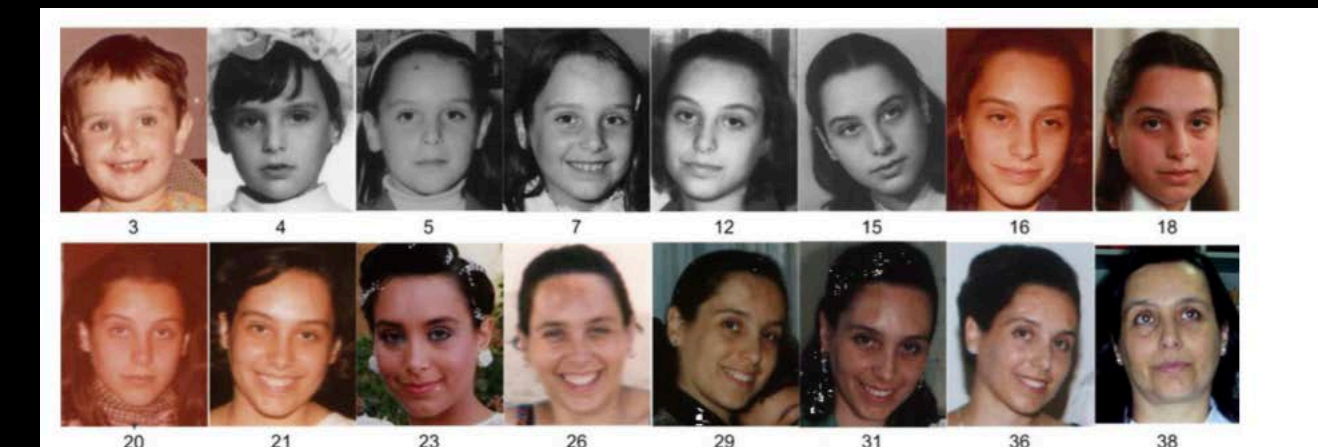
Face detection & tracking



Expression Analysis



Facial pose estimation



Face age estimation & aging

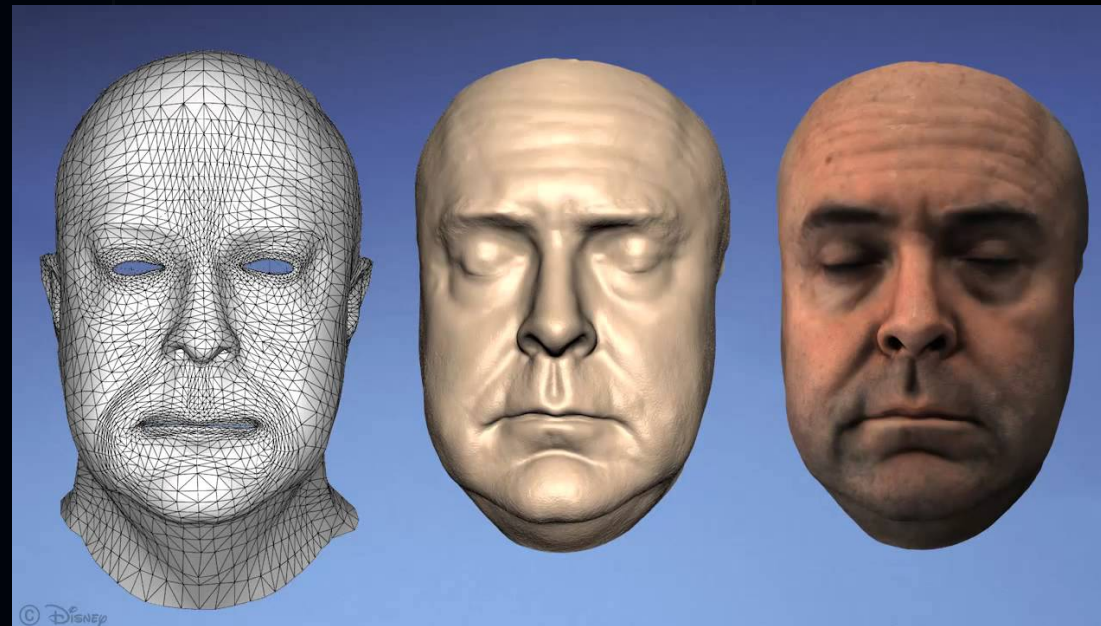
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Wide application of CV for human faces

Understanding human faces is fundamental for a variety of applications. It's not just about recognizing who the person is, but also about understanding their expressions, estimating their age, and even tracking their movements.



Face Reconstruction & Animation



Beauty Filter



Photo Galleries

[https://
machinelearning.apple.com
/research/recognizing-
people-photos](https://machinelearning.apple.com/research/recognizing-people-photos)

[https://
developers.google.com/static/
ml-kit/vision/face-detection/
images/face_contours.svg](https://developers.google.com/static/ml-kit/vision/face-detection/images/face_contours.svg)

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Face -

Detection

finding whether there are any faces in the image or not. If present, then it also returns the position of the face(s) within the image

Verification

It checks "Is this the claimed person?". For example, a system asking for facial verification scans the face and matches it with the stored image to confirm identity.

Recognition

It answers "Who is this person?". It recognizes the person by comparing the face with all the stored images and finds the best match.

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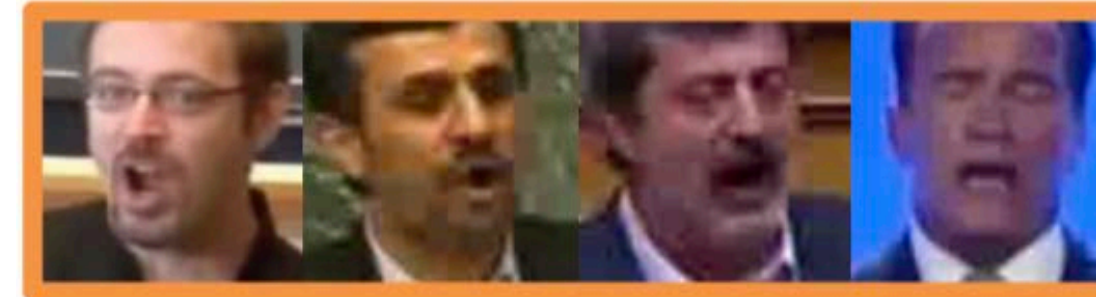
Challenge s

Face alignment and recognition come with their own set of challenges. These include dealing with large poses, varying facial expressions, partial occlusions, and the high-dimensional nature of the data we're working with.

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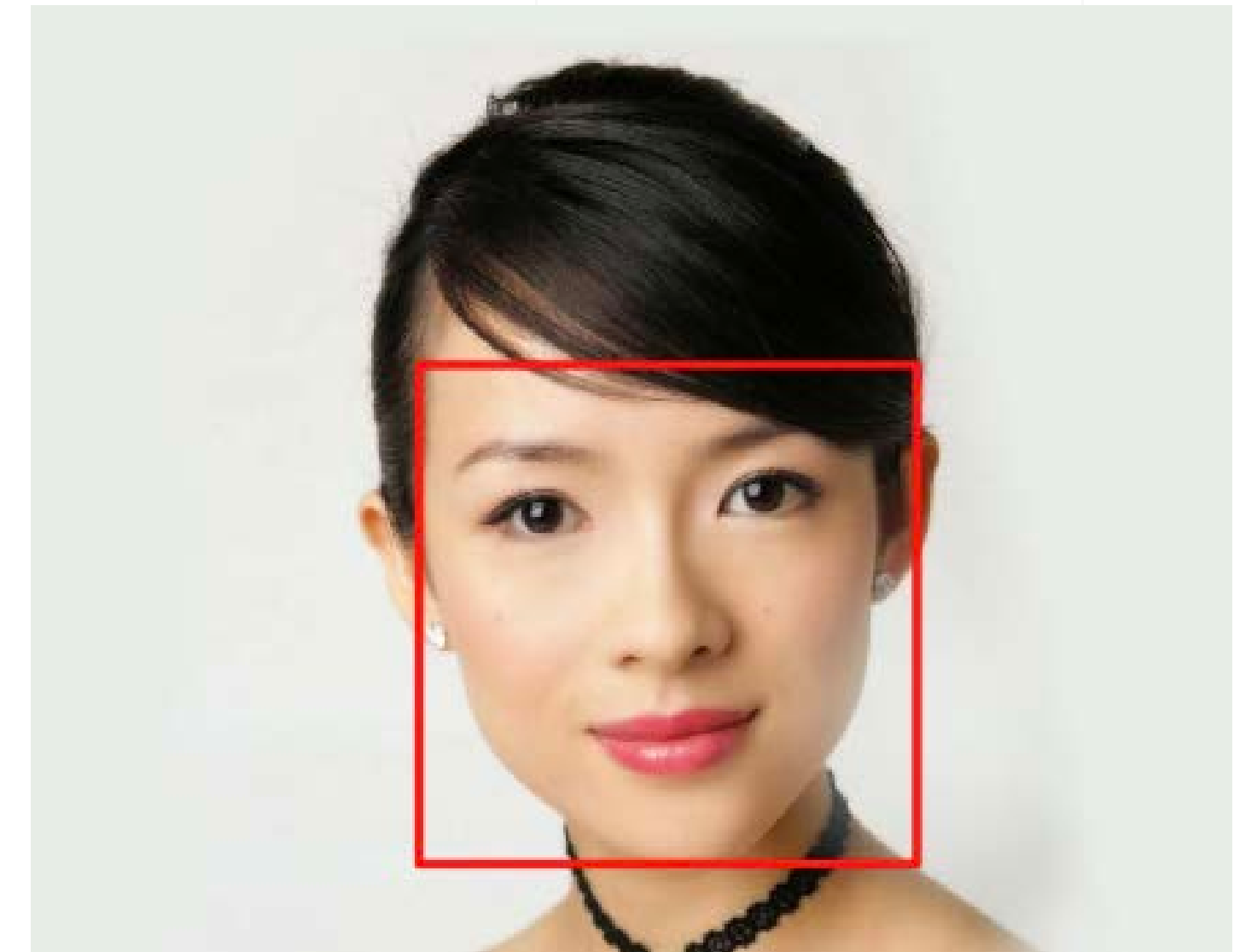
Large Pose



Various Expressions



Partial Occlusions





Face Analysis + Recognition Pipeline

FACE RECOGNITION PIPELINE



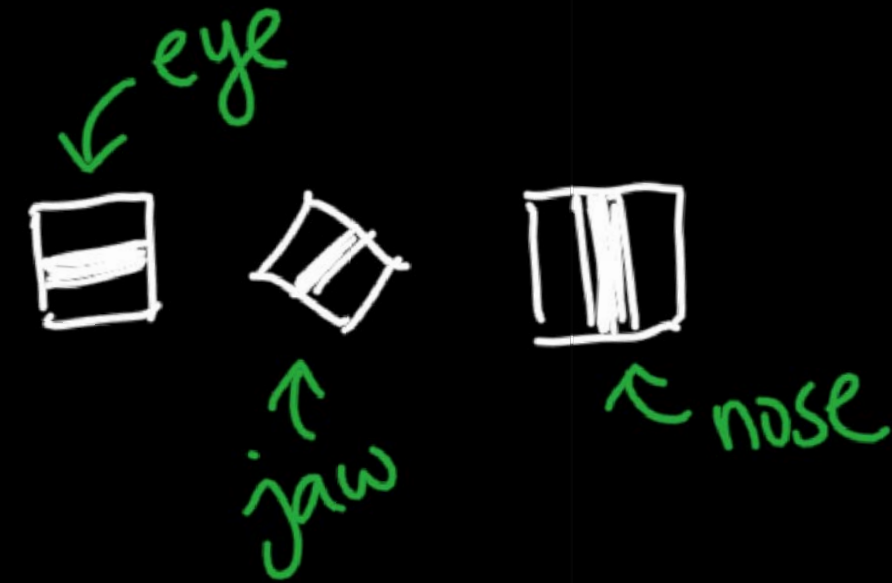
Detection

↳ Pixels

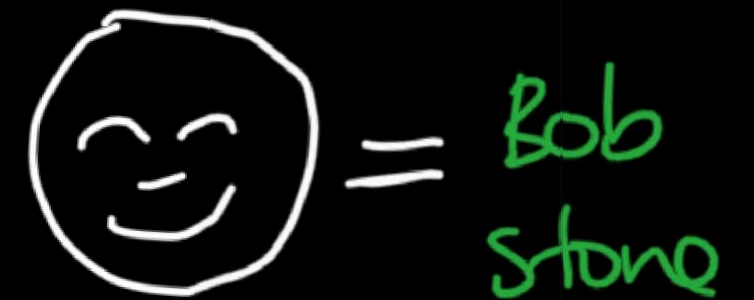


Alignment

↳ Facial landmarks



FEATURE EXTRACTION
OR MODEL LEARNING



RECOGNITION





Representation learning takes a different approach. Instead of manually designing features, we train our models to learn these features themselves. This can lead to more robust and effective models. There are different types of representation learning, including Holistic, Local, Structural, and Hierarchical. Each of these types has its unique approach and benefits.

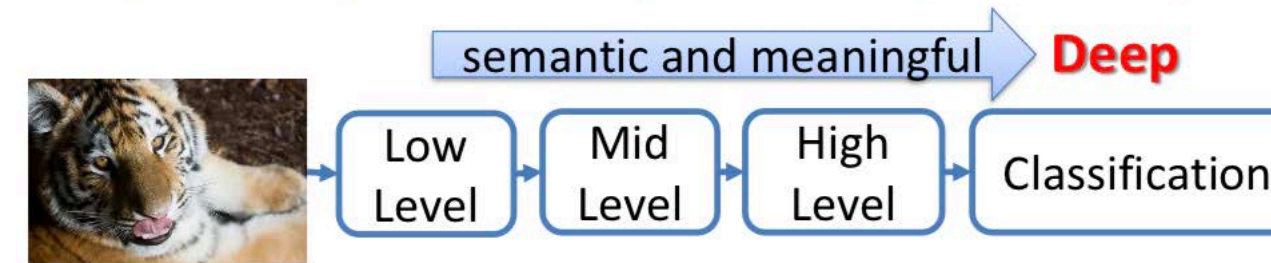
If you want to dive deeper into face analysis and recognition techniques I recommend following tutorial:

http://ivg.au.tsinghua.edu.cn/people/Yueqi_Duan/FG18_face.pdf

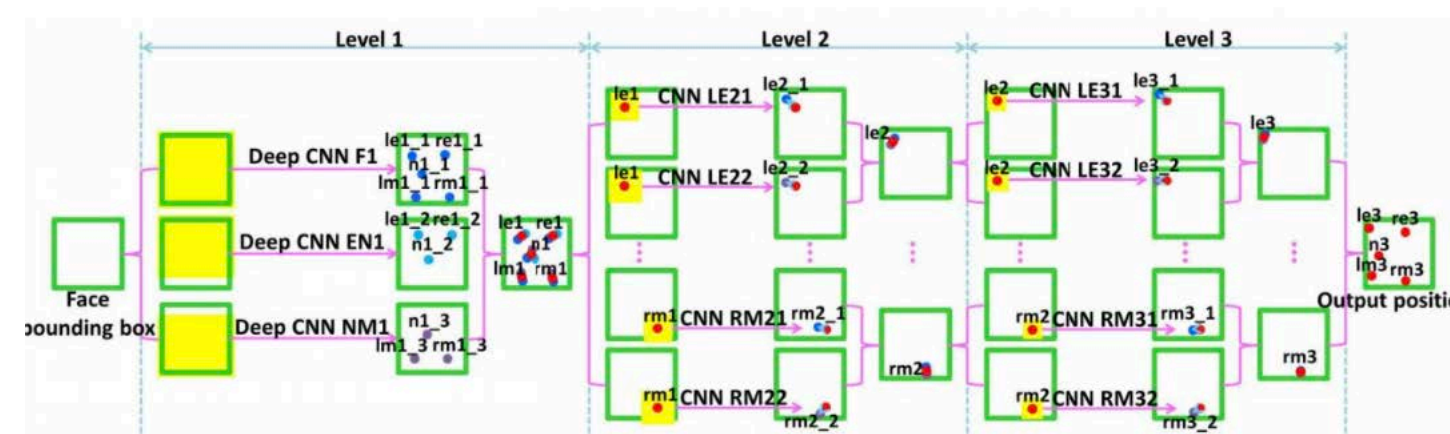


Hierarchical Representation: Deeply Learned Features

- Visual Recognition
 - pixel $\rightarrow$ edge $\rightarrow$ texture $\rightarrow$ pattern $\rightarrow$ component $\rightarrow$ object
- Text Classification
 - character $\rightarrow$ word $\rightarrow$ word group $\rightarrow$ clause $\rightarrow$ sentence $\rightarrow$ story



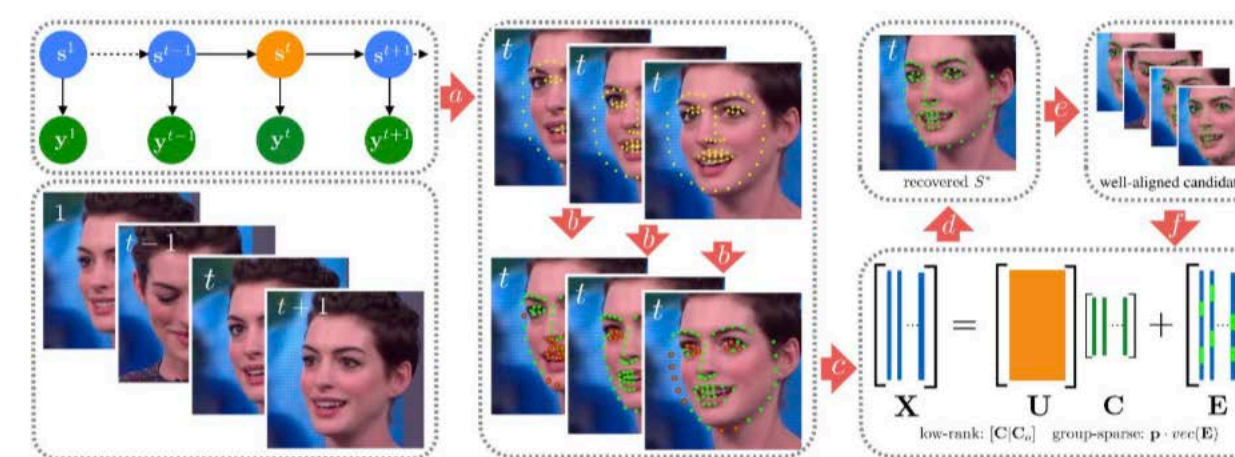
Learning Hierarchical Representation by Convolutional Networks



[Zhang et al., CVPR'13]

- Level 1
- Level 2 & 3
- Limitations

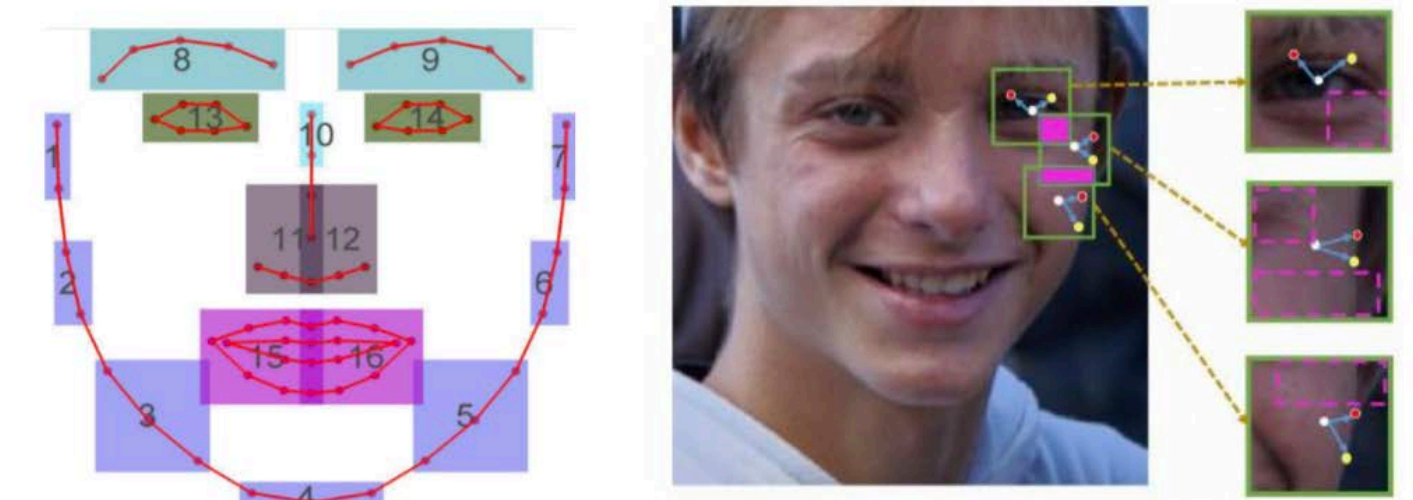
Learning Temporal Representation for Video-based Face Alignment



- Challenges for Video-based Face Alignment
 - Consistency over time steps
 - Robustness to initialization of both spatial and temporal dimension

Learning Deep Structural Representation

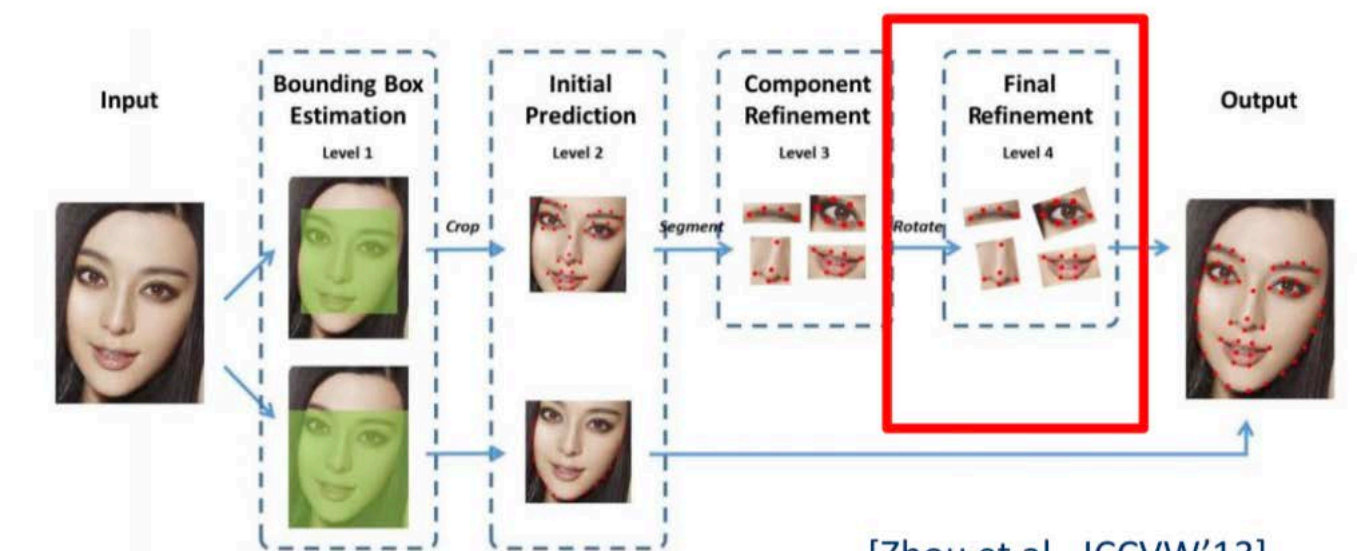
Motivation



Semantic Facial Parts

Structural learning from neighbouring landmarks

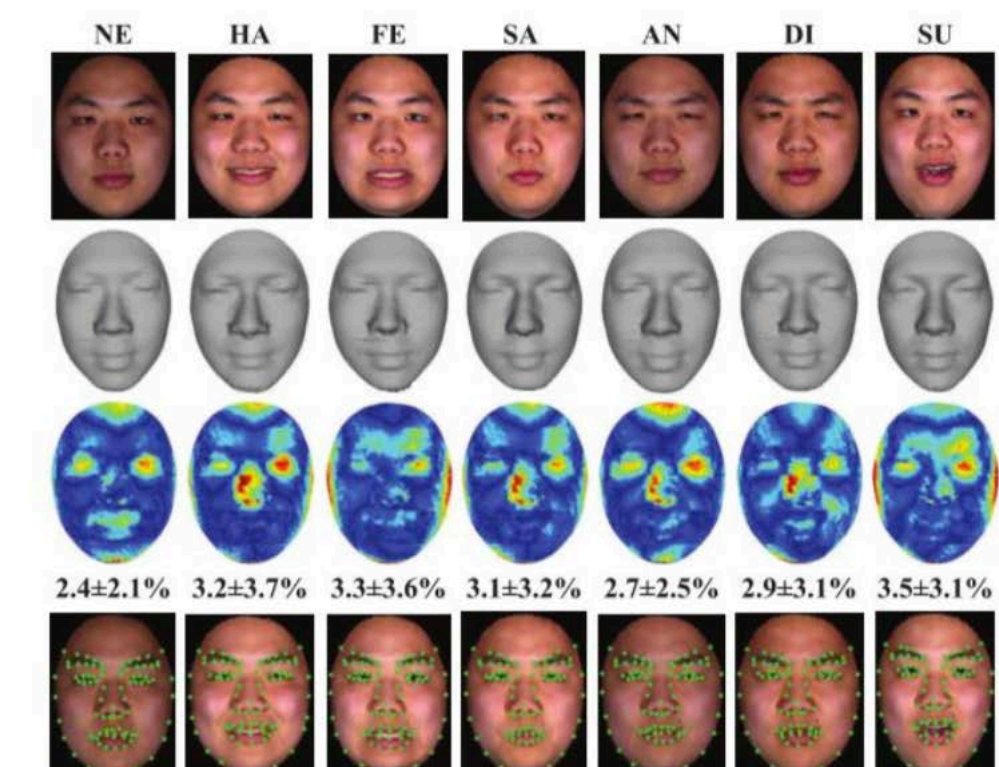
Learning Hierarchical Representation by Facial Landmarks Partition



[Zhou et al., ICCVW'13]

- Basic Idea

Learning Multi-Task Representation with 3D Surface Reconstruction





OpenCV

- `cv2.CascadeClassifier()`: This function is used for face detection. It uses a cascade of boosted classifiers working with haar-like features.
- `cv2.face.createEigenFaceRecognizer()`, `cv2.face.createFisherFaceRecognizer()`, `cv2.face.createLBPHFaceRecognizer()`: These are some of the methods used for face recognition in OpenCV.
- `cv2.getRotationMatrix2D()`, `cv2.warpAffine()`: These functions are used to align faces.

Dlib

- `dlib.get_frontal_face_detector()`: This function returns the default face detector. This function is equivalent to calling `dlib.frontal_face_detector()`.
- `dlib.shape_predictor()`: This function is used to predict face shape, which is then used to align the face (pre-downloadable model)
- `dlib.face_recognition_model_v1()`: This is used for face recognition.

Face recognition library

- `face_recognition.load_image_file()`: This function loads an image file into a numpy array.
- `face_recognition.face_encodings()`: Given an image, returns the 128-dimension face encoding for each face in the image.
- `face_recognition.compare_faces()`: Given a list of face encodings, compare them to a known face encoding and get a euclidean distance for each comparison face. The distance tells you how similar the faces are.

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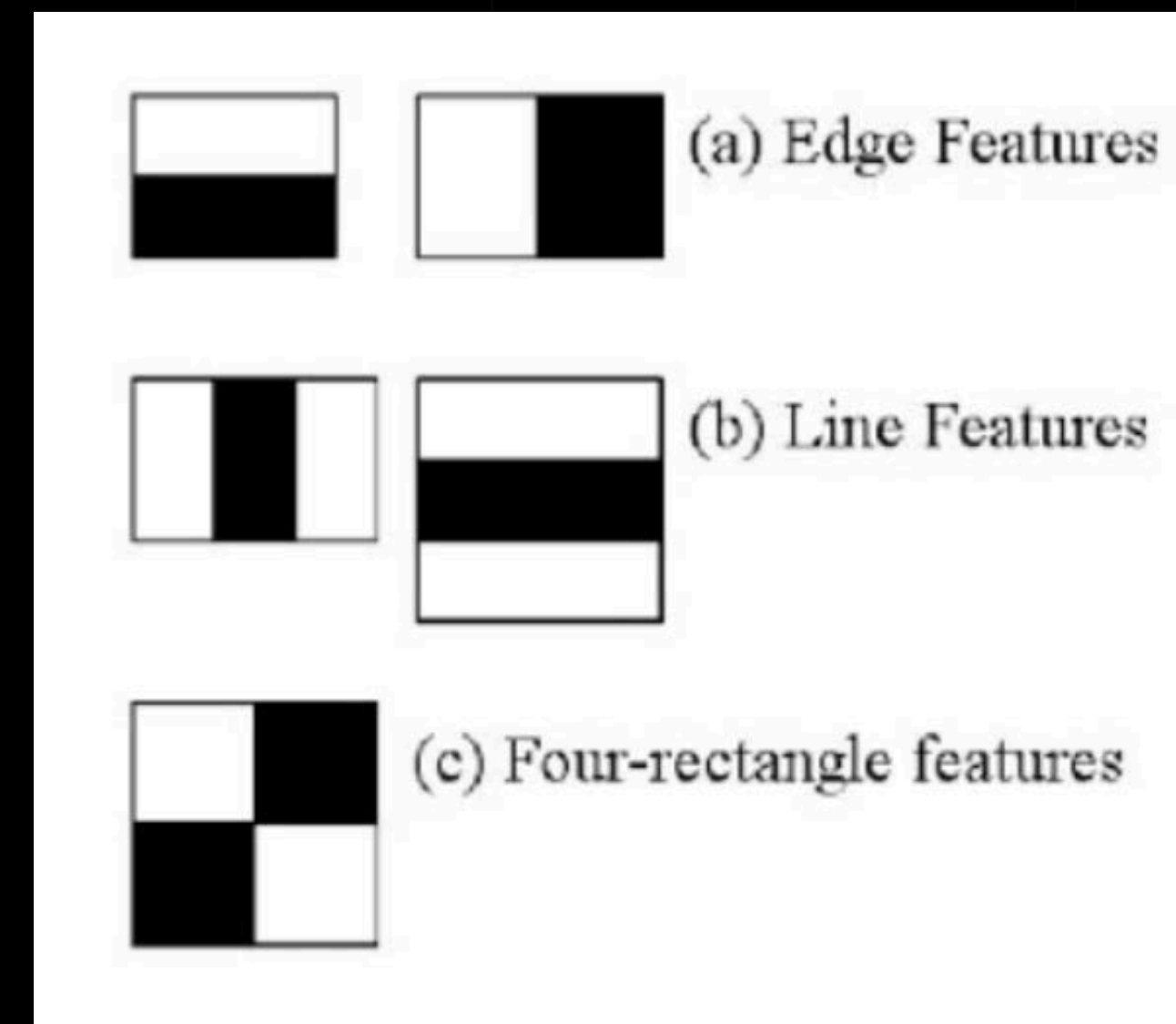
Haar Cascades

In 2001 Paul Viola and Michael Jones published a method of face detection based on the simple concept of a few key features.

They also came up with the idea of pre-computing an integral image to save time on calculations.

Main features: edge Features, Line features, Four-rectangle feature

Each feature is a single value obtained by the sum of pixels under a white rectangle from the sum of pixels under a black rectangle





Feature calculation

mean (dark region) - mean(light region)

A perfect edge would result in a value of one

sum of dark region

$([0.8,1,0.7,0.8,0.8,0.8,0.8,0.8]) = 6.5$

avg of dark region = $6.5/8 = 0.8125$

sum of light region

$([0,0.1,0.3,0.1,0.1,0.2,0.2,0.2]) = 1.2$

avg of light region = $1.2 / 8 = 0.15$

Delta = $0.8125 - 0.15 = 0.6625$

(indicates an edge)

The closer we are to 1 the better the feature (pass or no pass based on threshold).

0	0	1	1
0	0	1	1
0	0	1	1
0	0	1	1

0	0.1	0.8	1
0.3	0.1	0.7	0.8
0.1	0.2	0.8	0.8
0.2	0.2	0.8	0.8





Integral Images

- Calculating the sums for an entire image is very computationally intensive
- Viola-Jones algorithm solves this by using the **integral** images
- Resulting in an Order algorithm(1) running time of the algorithm
- Integral picture = Summed Area table
- Image 1 indicates the original picture with original pixel values; on the left is the integral image.
- It allows us to calculate the sum of any rectangle quickly; image; each colored spot highlights the sum inside the rectangle of that color, which is essentially
- This Integral Image is pre-calculated allowing this algorithm to run very fast

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1.

31	2	4	33	5	36
12	26	9	10	29	25
13	17	21	22	20	18
24	23	15	16	14	19
30	8	28	27	11	7
1	35	34	3	32	6

2.

31	33	37	70	75	111
43	71	84	127	161	222
56	101	135	200	254	333
80	148	197	278	346	444
110	186	263	371	450	555
111	222	333	444	555	666

$$15 + 16 + 14 + 28 + 27 + 11 = 111$$
$$101 + 450 - 254 - 186 = 111$$



Cascade Classifier

- Treat image to a series (a cascade) of classifiers based on the simple features shown earlier (edges, lines)
- Once an image fails a classifier, we can stop attempting to detect a face
- The algorithm is not slowly scanning the image looking for an image, it's much more efficient with pass/ no pass detection of features

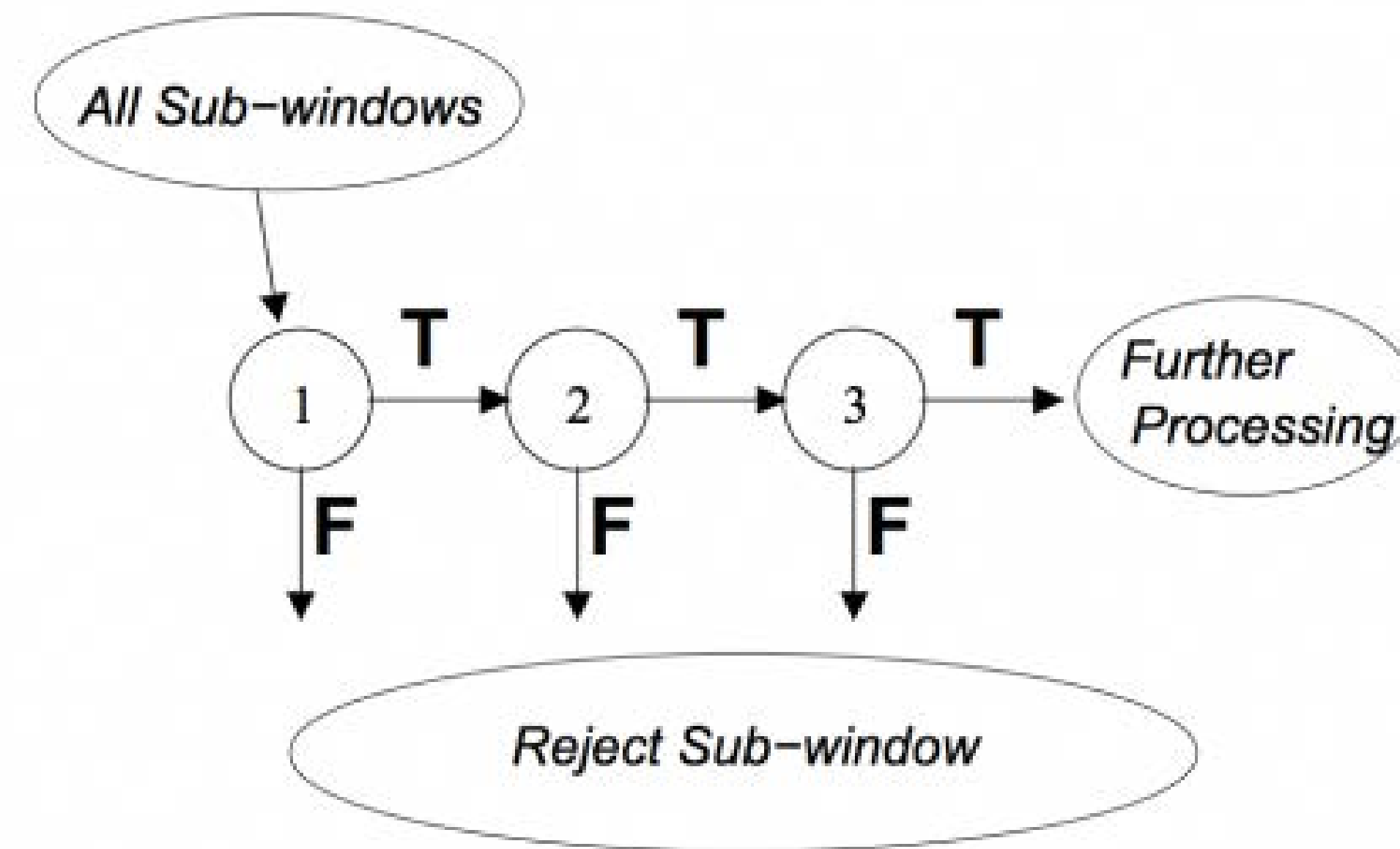


Figure 4: Schematic depiction of a the detection cascade.

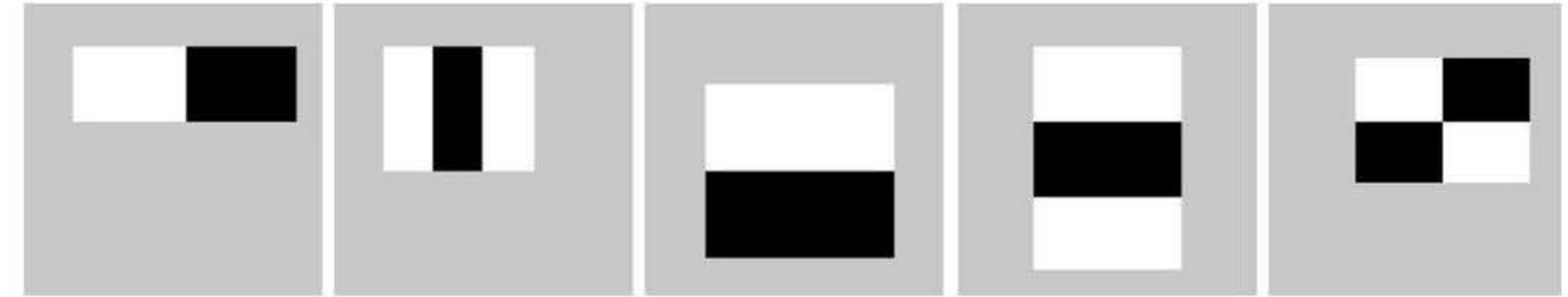




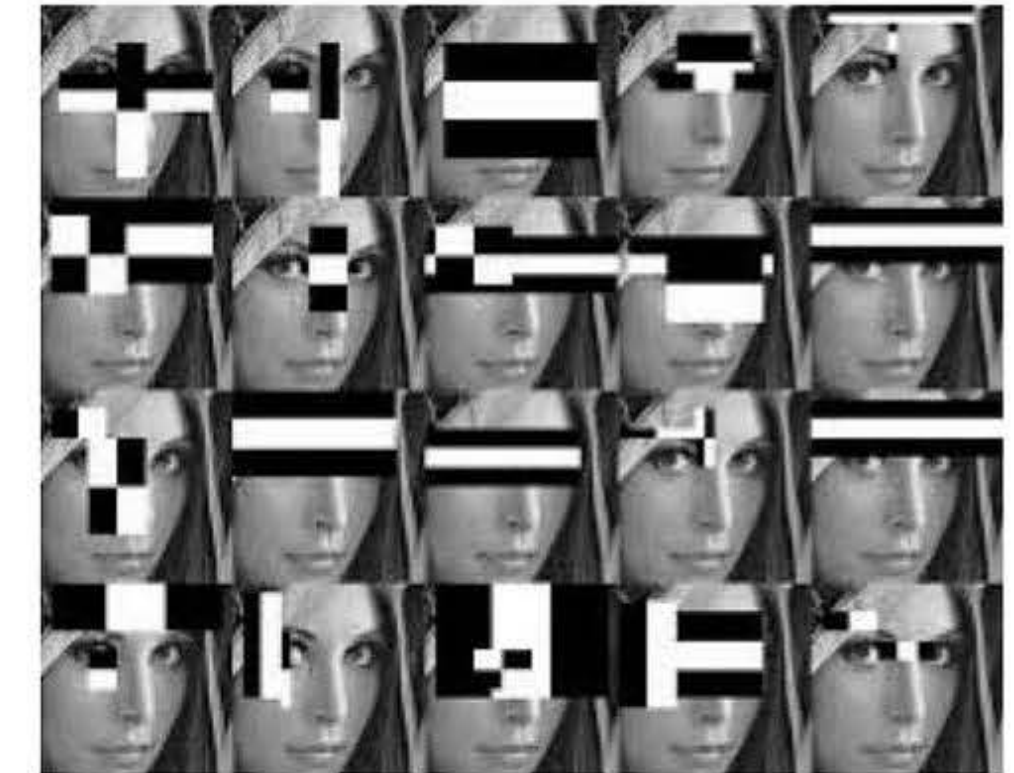
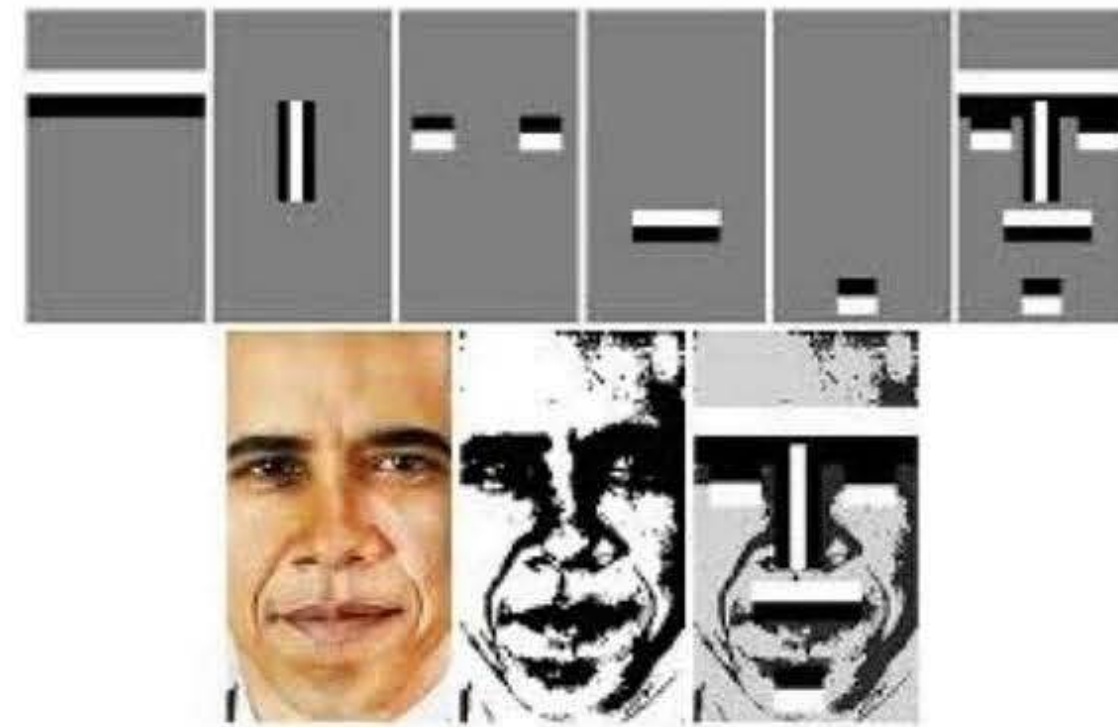
Finding Faces

- 1) Frontal Face of a person
 - 2) Greyscale image, because computer's only interested in light vs. Dark region
 - 3) Search for features (often line feature for eyes) // if cascade were to fail this it would stop already looking for a face
- Algorithm Challenges:**
- Very large data set required, but many pre-trained cascades exist, even for your cats.
 - False positives possible, and manual tweaking necessary

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(a) Edge Feature (b) Line Feature (c) Edge Feature (d) Line Feature (e) Four-Rectangle Feature





Practical Task - Exploring OpenCV

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Detecting Faces & Features

Goal:

Understand how classical Computer Vision detects features in grayscale images.

Your Task:

1. Choose one image from your dataset and load it in OpenCV.
2. Convert it to grayscale using an appropriate method.
3. Experiment with at least one Haar cascade (faces, eyes, smiles...).
4. Adjust detection parameters to improve results.
5. Visualize the detected regions and reflect: *When does it fail? Why?*

Extended Challenge 🧑🏫:

Adapt your approach so it works on a **video or webcam stream**.

Detect multiple features at once and discuss how stability changes across frames.

Reflection:

What influences detection performance most — the algorithm, the parameters, or the image itself?

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AI Bias & Racial Profiling in Face Detection

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Face Recognition is toxic

1. **Potential for misuse:** Face recognition technology can be misused for unlawful surveillance, invasions of privacy, or by governments to suppress dissent. It's crucial to have strong regulations and safeguards in place to prevent misuse.
2. **Bias and discrimination:** Research has shown that face recognition systems can exhibit bias, with higher error rates for women, the elderly, and individuals of certain ethnicities. This can lead to unfair treatment or outcomes if the technology is used in law enforcement or job recruitment.
3. **Consent and privacy:** Using someone's face for recognition without their consent can be seen as an invasion of privacy. Clear policies and mechanisms are needed to ensure people are aware of and consent to using their biometric data.



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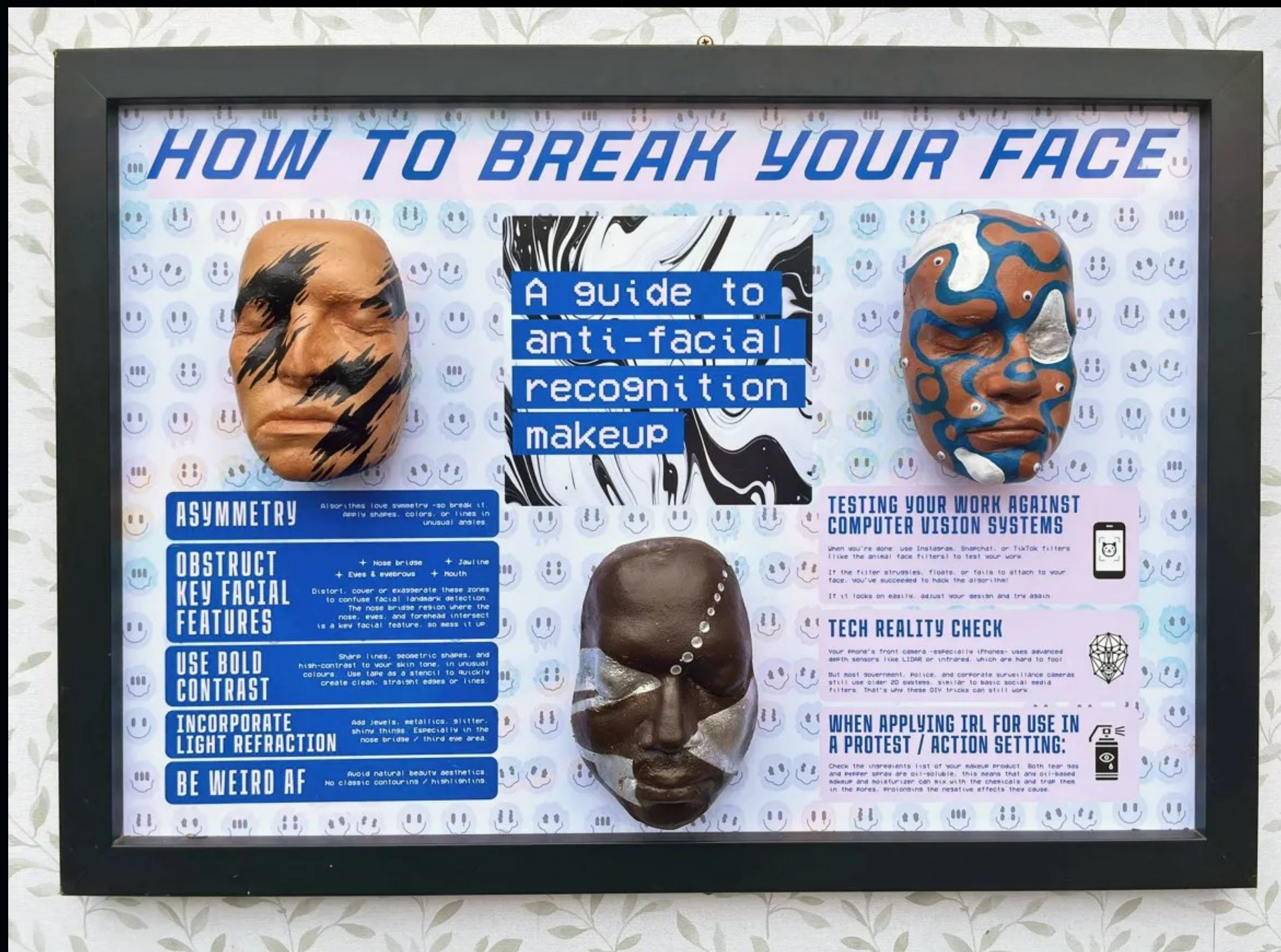




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Racial face recognition
zones?





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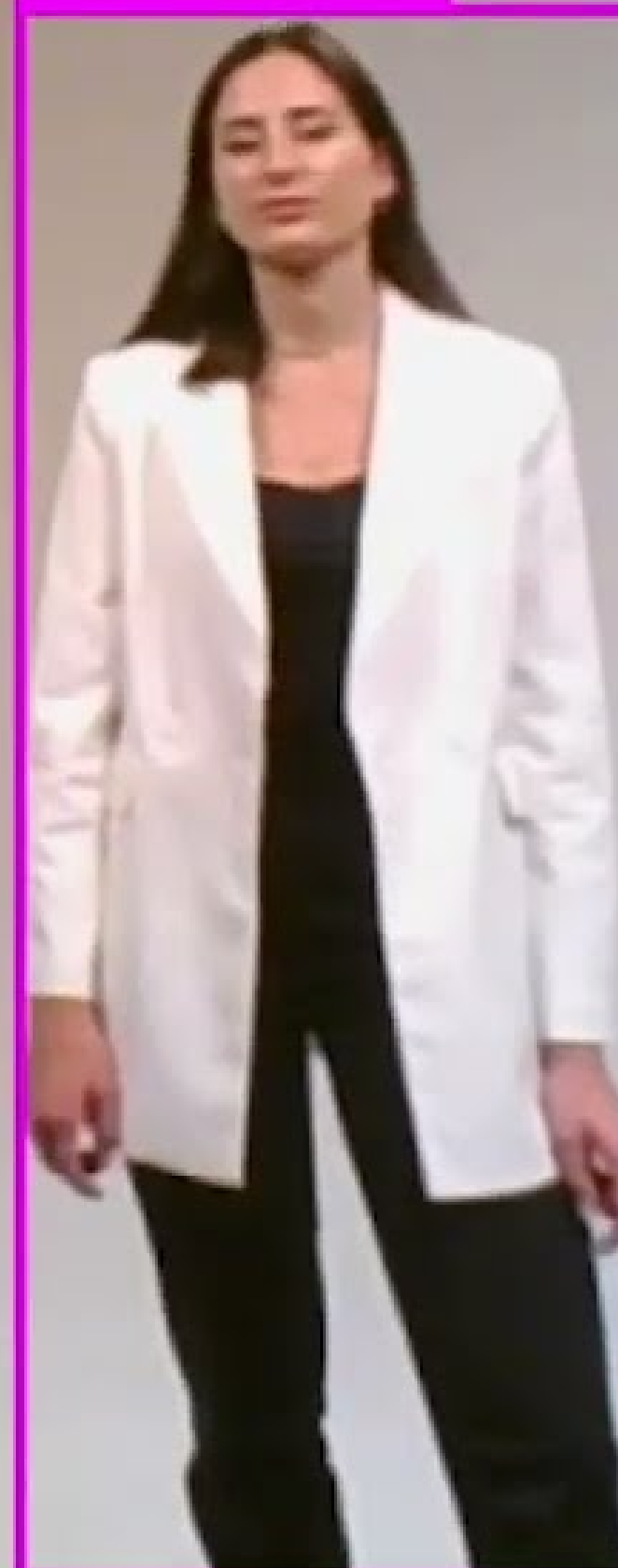




per son



per son



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IBM & AWS stance on face recognition

1. **IBM's decision:** In June 2020, IBM announced they would stop selling facial recognition services, citing potential misuse by law enforcement and violation of basic human rights and freedoms.
2. **Promotion of ethics in AI:** IBM's decision reflects their commitment to ensuring artificial intelligence, including facial recognition technology, is used ethically and responsibly.
3. **Call for regulation:** IBM has urged Congress to implement stricter regulations on the use of facial recognition technology to prevent potential misuse and violations of privacy rights.
4. **Continued research:** Despite stopping the sale of the technology, IBM continues to conduct research in visual and facial recognition technology with a focus on eliminating bias and promoting fairness and accuracy in AI.

Apart from IBM also, AWS has stopped selling its Rekognition technology as a response to police brutality.



AI Ethics & Trustworthy AI is critical to the application of face recognition

Explainability: We provide tools to understand how AI systems make decisions and recommendations, enabling transparency and trust.

Bias Detection and Mitigation: We identify and mitigate bias in AI models and datasets to prevent unfair outcomes.

Privacy: We protect personal data and ensure compliance with data protection regulations.

Our AI Ethics Board reviews and advises on our AI initiatives' ethical and societal implications.

IBM is a founding member of the Partnership on AI, a cross-industry effort to promote AI that benefits society.

<https://www.ibm.com/watson/assets/duo/pdf/everydayethics.pdf>

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Everyday Ethics for Artificial Intelligence

Try how you test your data for bias with AIF360, an open-source toolkit that can help you examine, report, and mitigate discrimination and bias in machine learning models throughout the AI application lifecycle. We invite you to use and improve it:

<https://aif360.mybluemix.net>



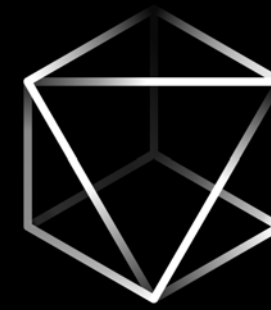


Kick Off Questionnaire

Thanks & Questions?

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